MASTER OF BUSINESS ADMINISTRATION

SEMESTER 3

DADS304

VISUALIZATION

Unit 14

Visualization with Power BI Desktop

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1. INTRODUCTION

Power BI Desktop is a Windows application that enables users to create advanced data visualizations and business intelligence reports using data from a variety of sources. It is a free, standalone desktop application that is part of the Power BI suite of business intelligence tools developed by Microsoft. Power BI Desktop allows users to create custom data visualizations, reports, and dashboards with an intuitive drag-and-drop interface. It includes a range of built-in data connectors for a variety of data sources.

1.1 Learning Objectives

At the end of this unit, you will be able to

- *Describe the various option available on a Power BI Desktop User Interface.*
- *Explain the process to integrate data sources into Power BI Desktop.*
- *Elaborate on the process to create visualizations of the Data.*
- *Explain the steps to filter and format the various graphs created.*

INSPIRED BY LIFE

2. POWER BI DESKTOP – FEATURES

Power BI Desktop is a free software that can be downloaded and installed from the Microsoft website. Once the app is installed and launched, the UI similar to Figure 1 appears.

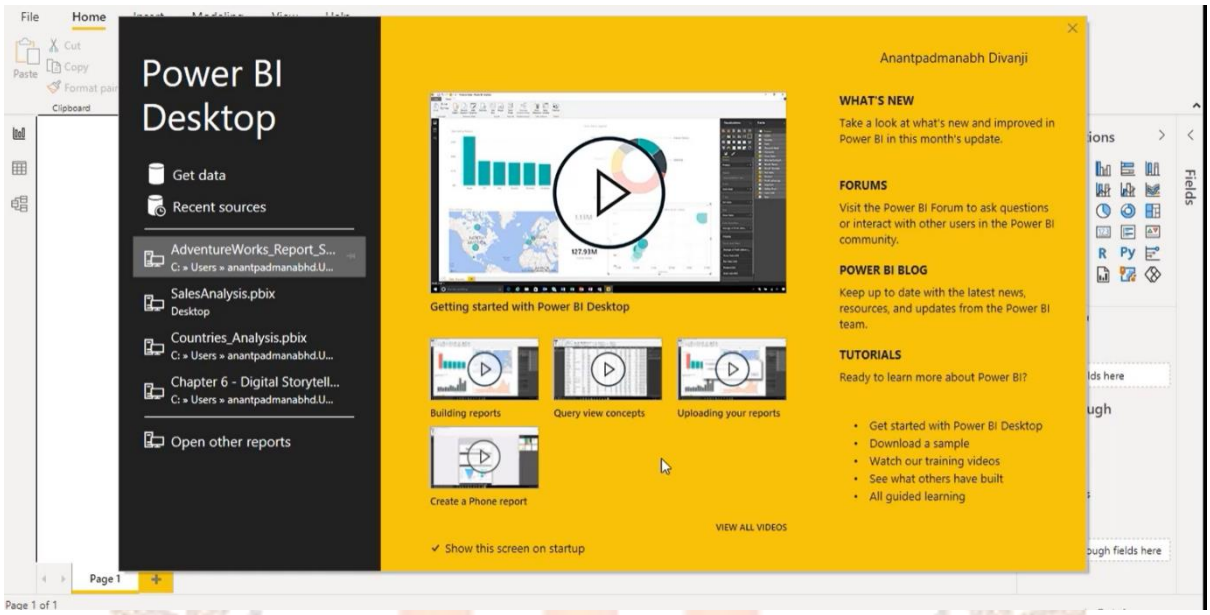


Figure 1: Power BI Desktop launch page

The start-up page or the launch page contains various sections. The 'Get Data' or 'Recent Sources' can be used to connect to a data source or import the data from any file. The files or reports that were previously worked on are listed as a *'quick link'* option. As a part of the start-up, links to tutorial videos are displayed. For example, there are videos on "*Getting started with Power BI Desktop*', 'Building Reports' and so on.

Power BI updates are released every month. The details of the latest update are also available as a link on this screen under "*What's new*". Besides this, the start-up page has a link to Forums, Blogs and Tutorials.

These options are always shown at every launch of the app. To not display them at start-up, the option "*Show the screen on the start-up*" can be disabled.

Once this launch page is closed, the main interface appears as shown in Figure 2.

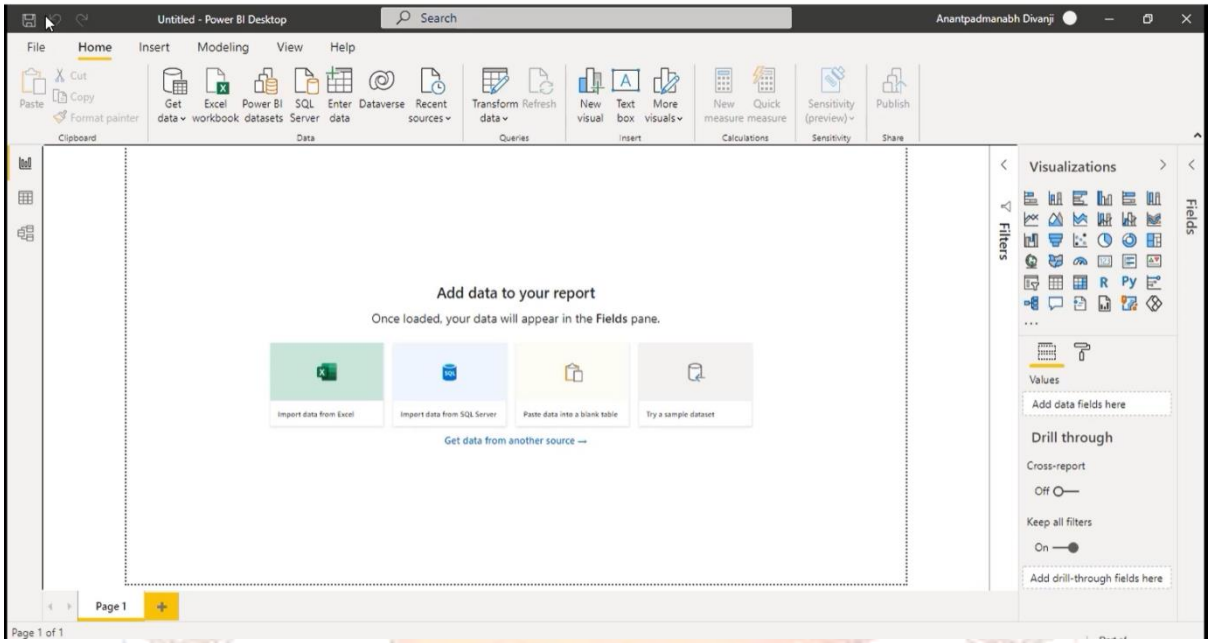


Figure 2: Main User Interface

The main menu on the UI contains the following options

1. File Operation
2. Main Menu
3. Insert
4. Modelling
5. View
6. Help

3. REPORT CANVAS

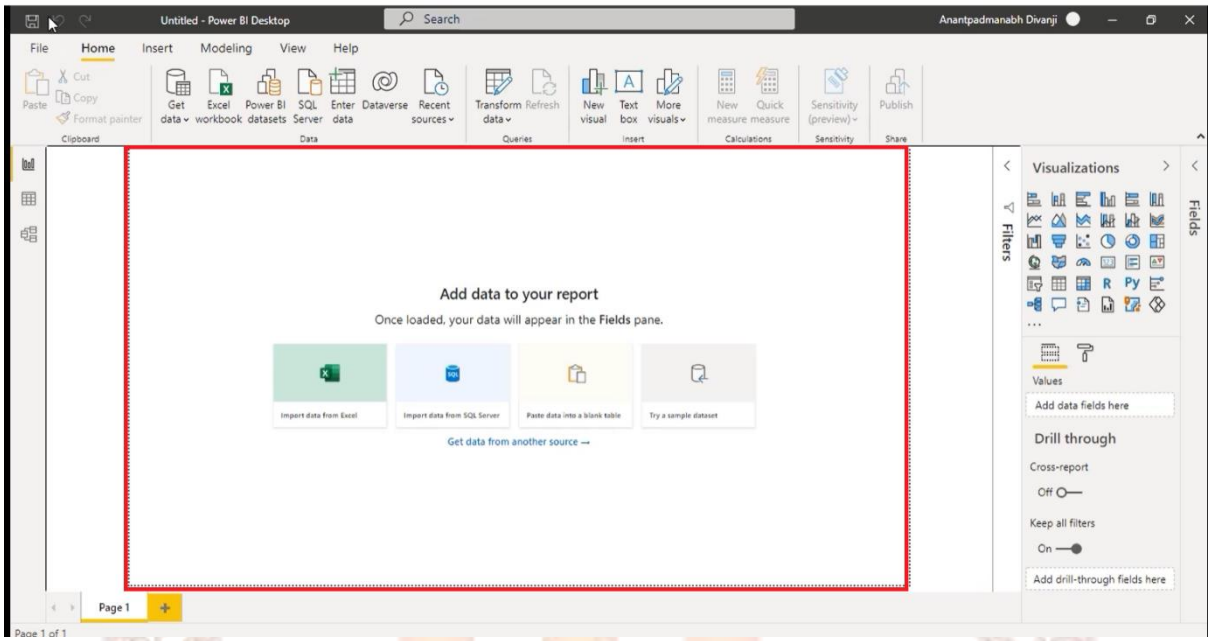


Figure 3: Report Canvas

The report canvas in Power BI is the main work area where users can create and design reports and visualizations using data from various sources. It is the area where users can drag and drop fields from their data sources, create charts and tables, and arrange visualizations to create a cohesive report. Users can add new pages to their reports, and each page has its own report canvas. The canvas can be customized to fit the user's needs, including adding background colours or images and changing the page size.

The canvas in Power BI Desktop includes several layout tools that allow users to position and resize their visualizations. It also includes a formatting pane that allows users to adjust the formatting and appearance of their visualizations, including font size, colours, and other formatting options.

Pages can be added and deleted on the canvas using the '+' option. Along with the report canvas, there is an option to add filters. This can be applied on a single page or on all pages. The page also contains standard or pre-defined visualisations that can be added to the main canvas.

Once the connection is established with a dataset, the columns corresponding to the datasets are available under fields. This can be dragged and dropped into the respective fields to populate the chart on the canvas.

The UI also has options to display the data model view and the relationship view. The data model is used to preview the data. The relationship view displays the common relationship between tables and the common columns.

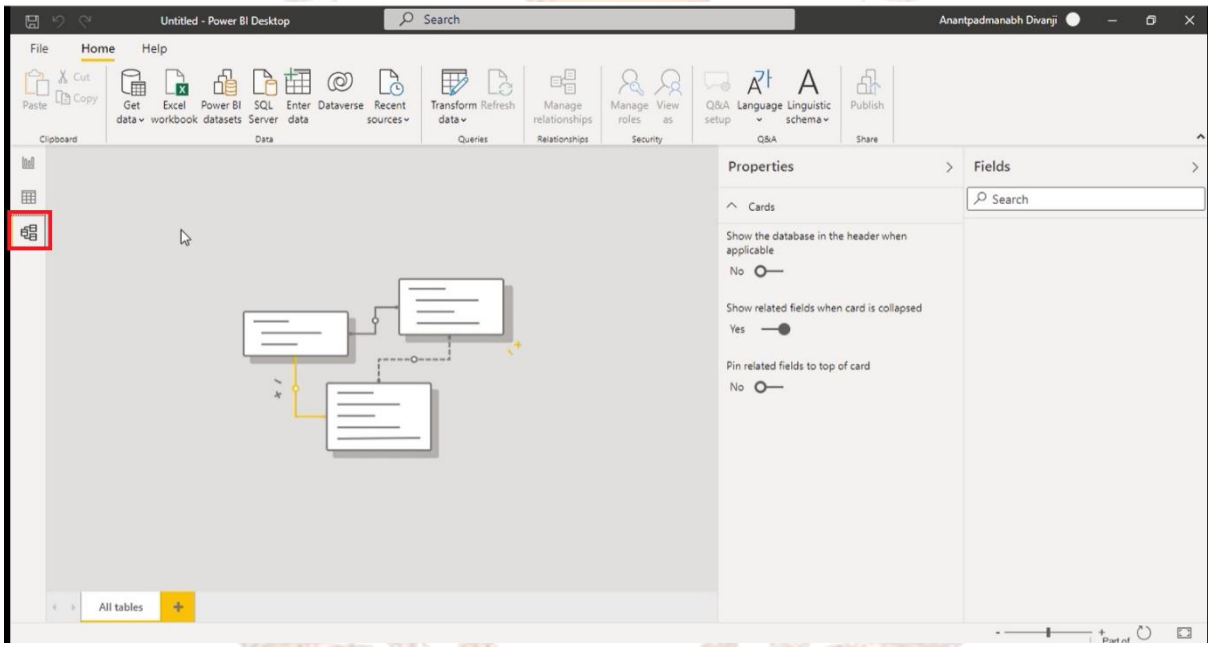


Figure 4: Relationship view

4. EXPLORING THE REPORT CANVAS

The report view in Power BI is where you design and build the visualizations and reports that you want to present to your audience. It offers a range of data visualization choices that may be used to present your data in a meaningful way, and it enables you to construct custom dashboards and reports using a drag-and-drop interface.

The workspace where you can design and generate your report in Power BI is called the report canvas page. It is the primary location where you can format the report layout, add and arrange visualizations, and apply filters. The canvas effectively serves as a blank page where you can start from the beginning when creating your report and include any visualizations you require, including maps, charts, tables, matrices, and images.

To launch the report canvas, click on the 3 dot that is present towards the right-end corner of the page and click on remove to remove the content/object on the page (if any) as shown in figure 5.

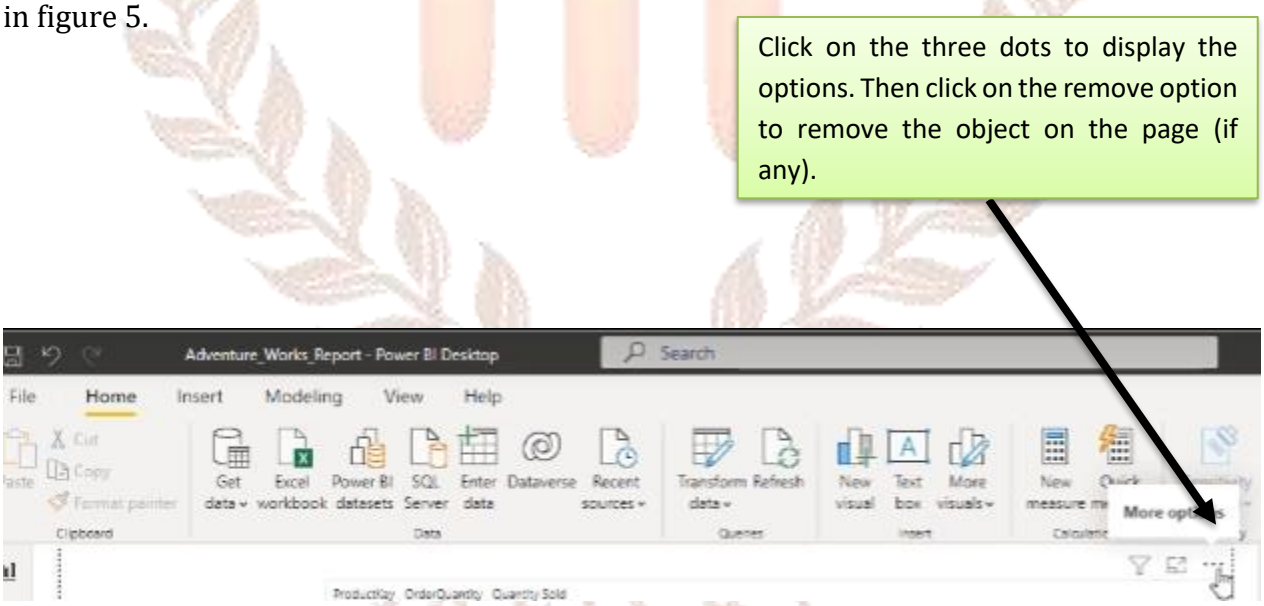


Figure 5: Option to remove the object on the page

Figure 6 shows the basic report canvas that we get at the start when we don't have any object placed on the report canvas (it's just an empty canvas).

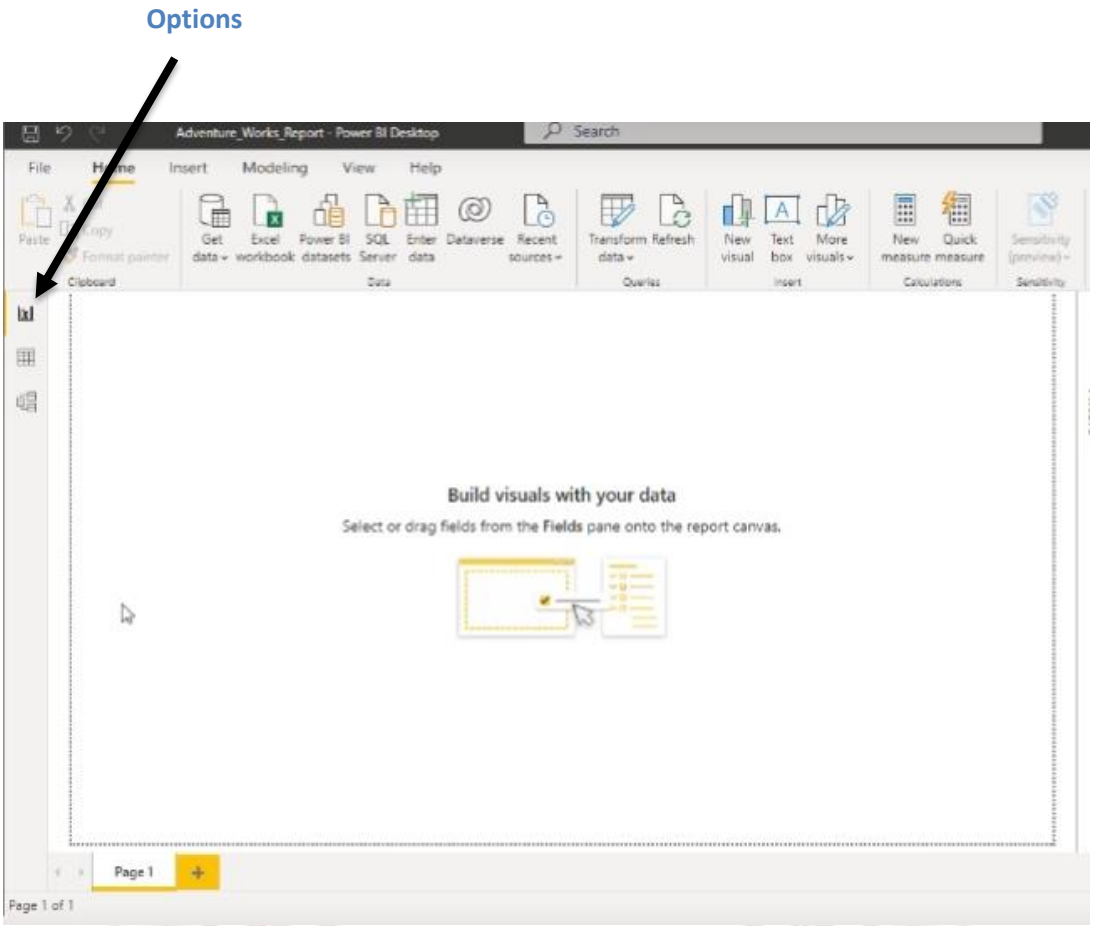


Figure 6: Empty/New Report Canvas

We have report canvas options and basic menus (file, home, insert etc.) as well on this page.

To change the background of the report canvas, click on view and change the background or themes of the report canvas as shown in figure 7.

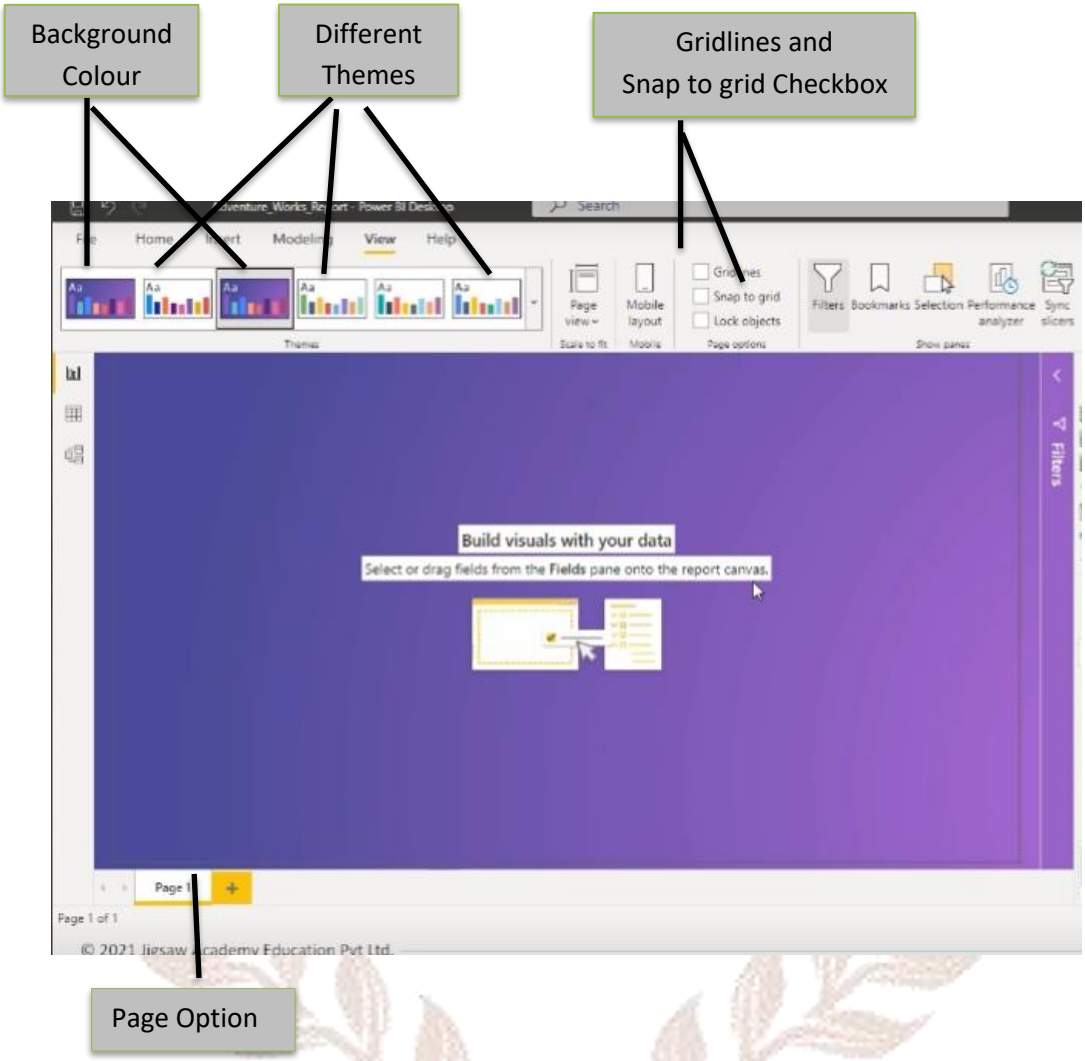


Figure 7: To change the background or theme of the report canvas

Then we have check boxes for

- Gridlines – The user may enable or disable the gridlines that show up in the report canvas by checking the "Gridlines" checkbox in the report canvas settings.
- Snap to Grid - You can enable or disable the snap-to-grid capability in Power BI's report canvas settings by checking the "Snap to grid" checkbox. When you move or resize visualizations with snap-to-grid enabled, they will automatically align to the closest grid line.

Some of the key features and components of the report view are:

- **Pages:** A report in Power BI can have one or more pages. The "New Page" button can be used to add new pages, and the page navigation bar at the bottom of the screen can be used to switch between pages as shown in figure 7.
- **Fields:** This section is where the actual data sets or data tables will appear. The user may view and manage each data field that is present in your dataset in the Fields pane. You can add fields to already-existing visualizations or create new ones by dragging fields from the Fields pane onto the canvas as shown in figure 8.
- **Visualizations:** These are the graphs, tables, maps, and other types of data visualizations that Power BI lets you make. A new visualization can be added to your report by choosing it from the Visualizations window on the right side of the screen, selecting it, and then dragging it onto the canvas as shown in figure 8.

In this visualization, we have a *Field view pane and a field formatting pane.*

- ✓ Field View Pane: The values in the fields list will be changing based on the visualization that we select.
- ✓ Field Formatting Pane: If a user clicks on the formatting option, the user can see the page-related information like Page Size, Page background etc. as shown in figure 8. you can expand them by clicking on the dropdown arrow and you can see the options for each field as shown in figure 9.

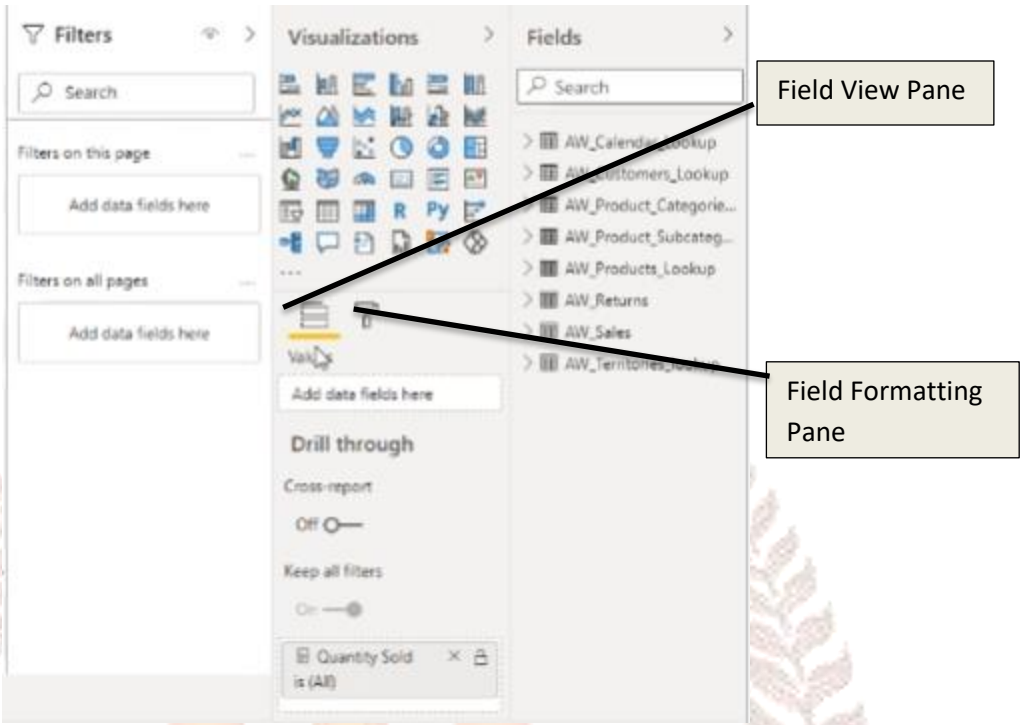


Figure 8: Showcasing the Fields, Visualization and Filters pane

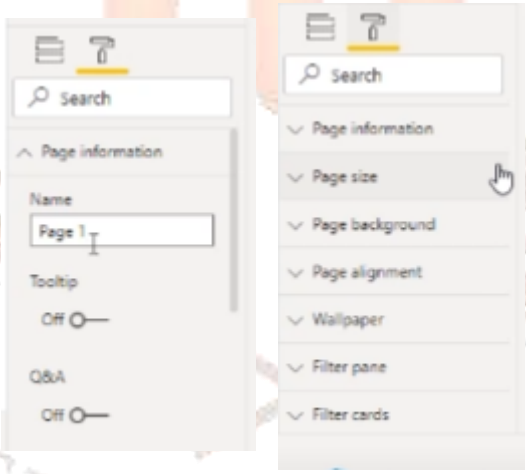


Figure 9: Options of the field formatting pane

- **Filters:** You may customize which data is shown in your visualizations by using filters. When creating a report, choose the visualization you want to filter, then click the "Filters" button in the Visualizations pane as shown in figure 10.

To the left of the Visualization section, we have Filter page/Section which is used to filter your visualization based on certain conditions. On this page, we have 3 options available i.e.,

- ✓ Filters on this page: To apply filters on the particular page
- ✓ Filters on all Pages: To apply filters on all the pages
- ✓ Filters on Visual: This field will be enabled when you add any visualization to the page and this filter can be applied to that particular visualization.

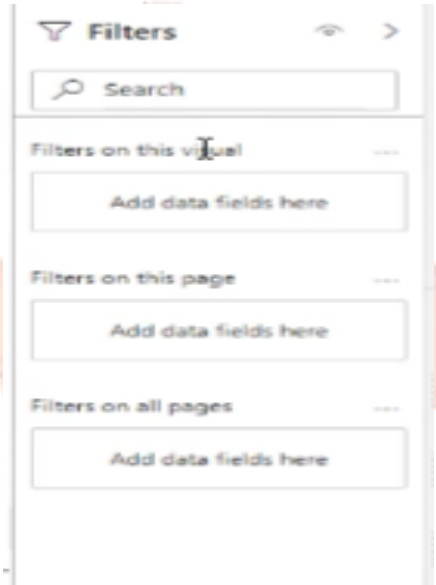


Figure 10: Options for Filters

5. BASIC CHARTS AND CONDITIONAL FORMATTING

For the demonstrations in this unit, the data from "Adventure Works" dataset is used. AdventureWorks is a free sample database of retail sales data.

Once the data has been imported, the columns (both inherent and derived) are available on the right of the Power BI Desktop view as shown in figure 11. The different charts available are also displayed here.

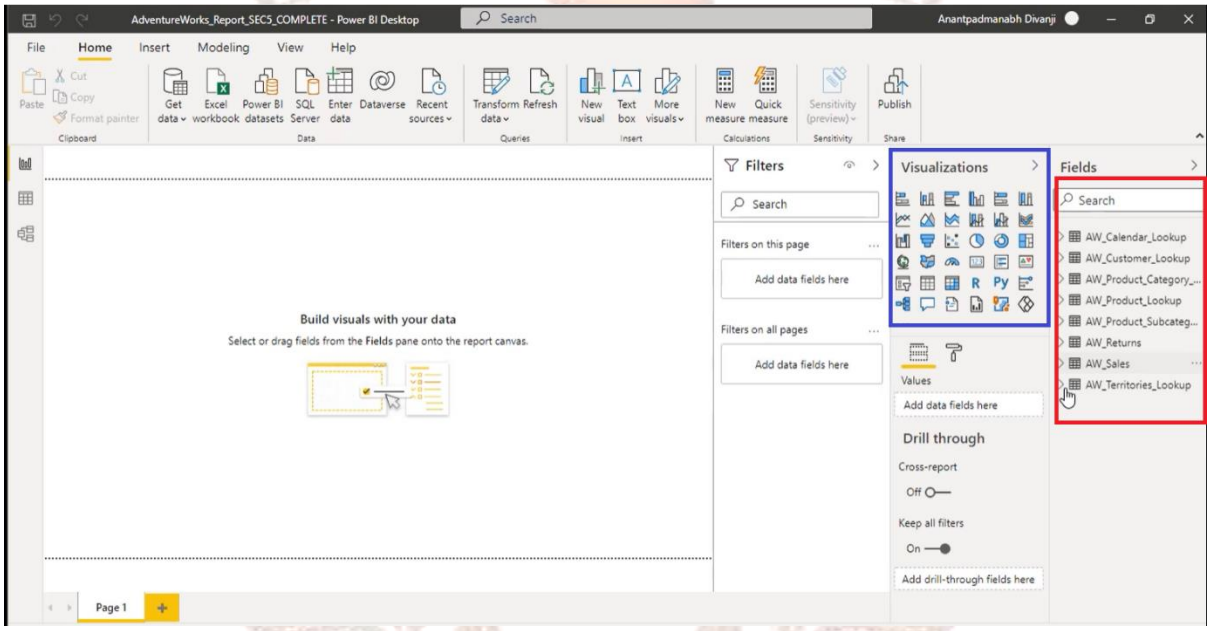


Figure 11 : Visualisation options and Fields on Power BI Desktop

Once a column is selected, the most appropriate chart corresponding to it is displayed on the view. For example, if the column 'Total Orders' is selected, a bar graph is selected and displayed on the screen. The attributes of the bar plot are available on the screen as shown in Figure 12.

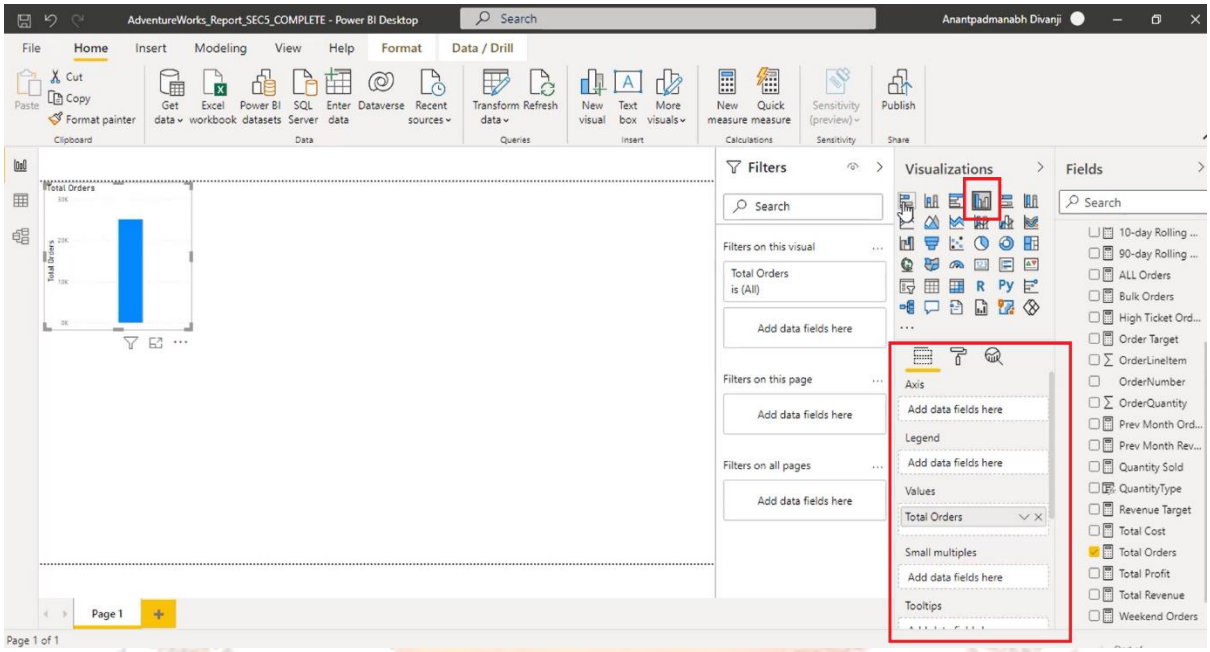


Figure 12: Bar plot with attributes

The type of graph can be changed by selecting the most appropriate one in the visualisation section. For instance, if the 'stacked bar chart' is selected, the graph will be converted into a horizontal chart. The attributes displayed will be that of a 'horizontal bar chart'.

We can create a new chart by selecting the 'New Visual' option as shown in Figure 13. This will create a default column chart.

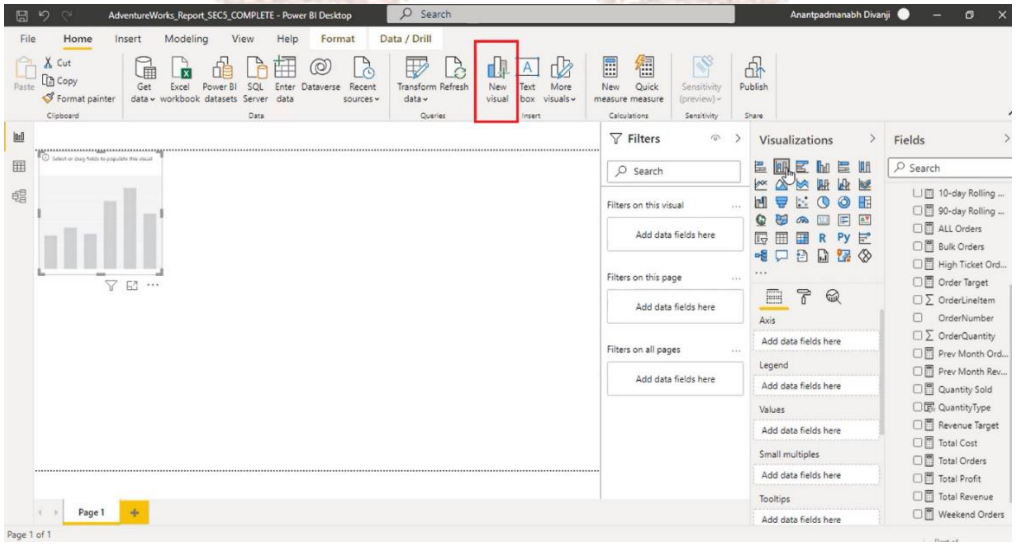


Figure 13: New Visual

The third method of creating a chart is by double-clicking on the canvass. This will create a default 'Q&A' chart.

Let us go back to selecting the 'Total Orders' as shown in Figure 14. The text corresponding to the 'Total Count' can be formatted using the options as shown in Figure 14. The number can be displayed as a comma-separated value by selecting the appropriate values.

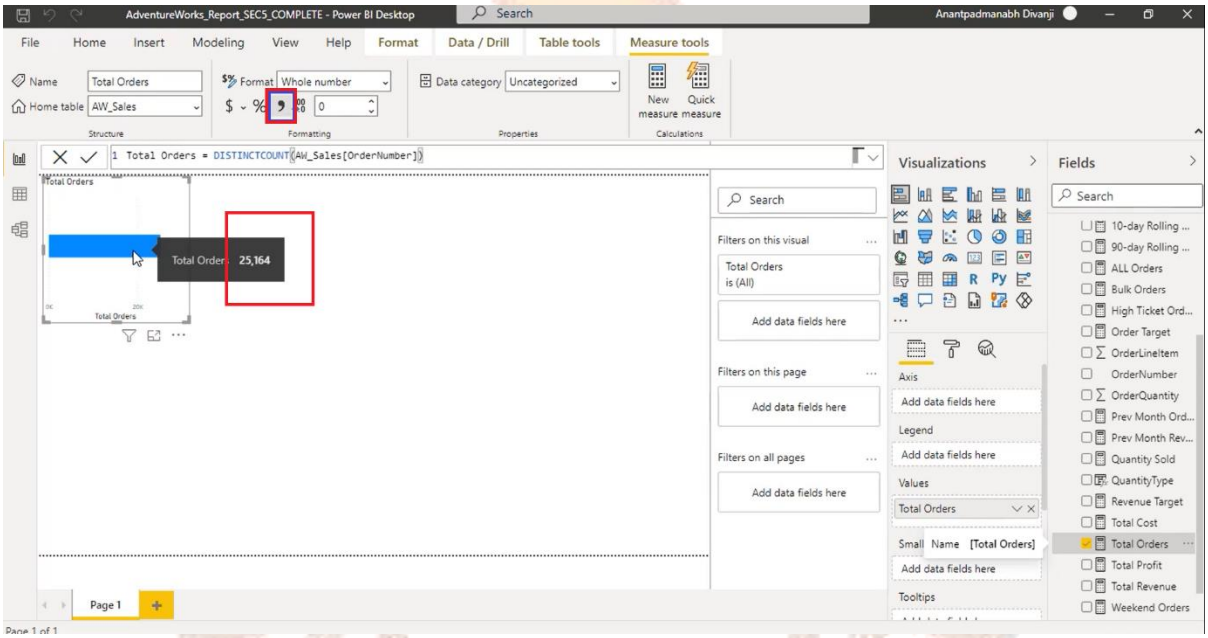


Figure 14: Formatting the text

To display the total orders subdivided by the subcategory names, 'select' the sub-category name from the 'fields' option and drag and drop into the 'Axis' attribute as shown in Figure 15. This will display the image as shown in Figure 15.

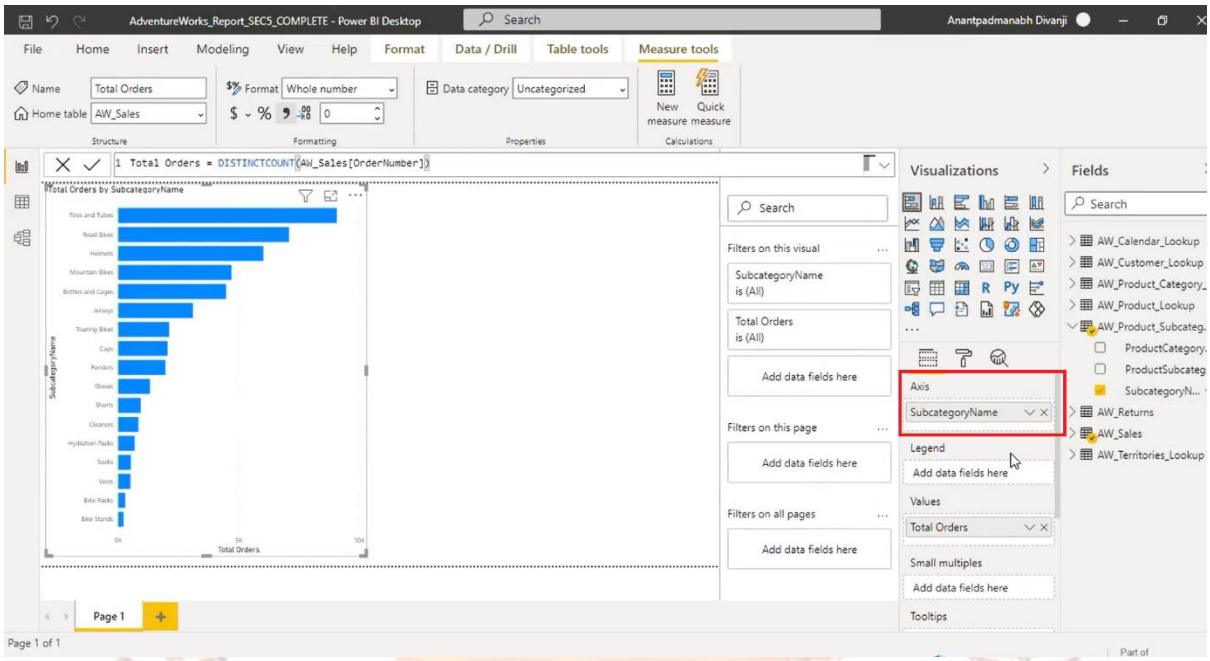


Figure 15: Bar plot per sub-category

Note that there is another attribute named 'Legend'. If the column 'Subcategory name' is moved to an attribute named 'Legend', the chart as shown in Figure 16 appears. This is not very easy to interpret.

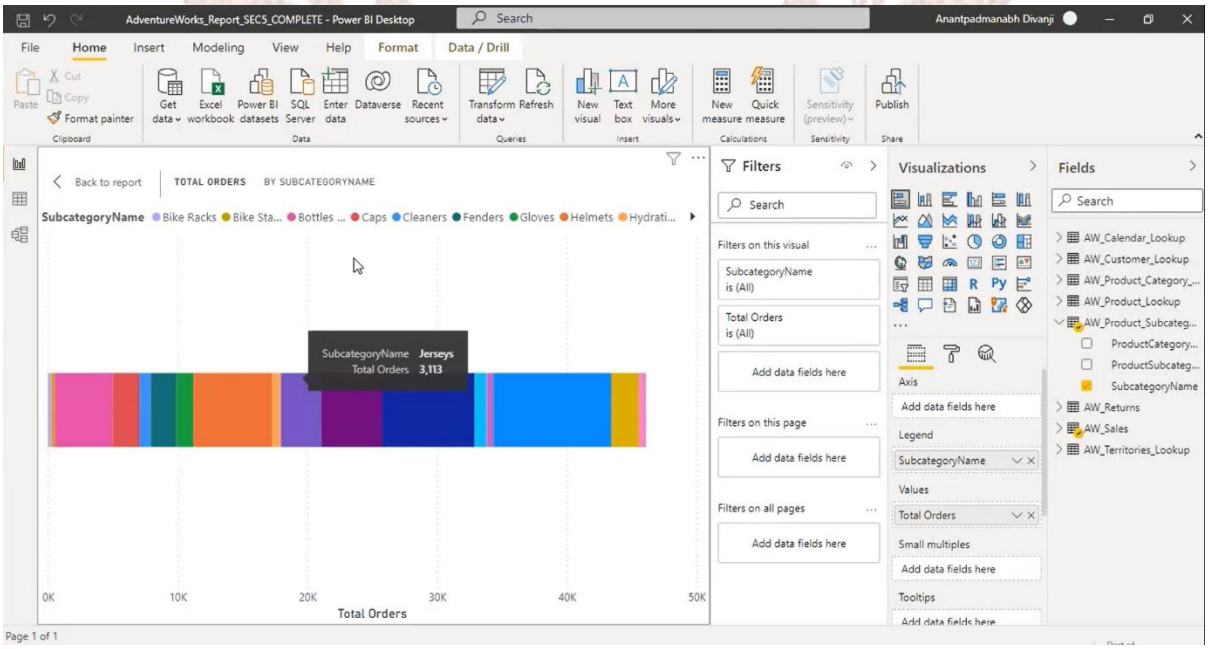


Figure 16: Bar plot with Subcategory as a 'Legend'.

The chart can be highlighted by clicking on the option 'Focus Mode'. The type of the chart can be modified by clicking on the 'Pie chart' option as shown in Figure 17. Note that the Attributes corresponding to the bar chart are also updated on the screen.

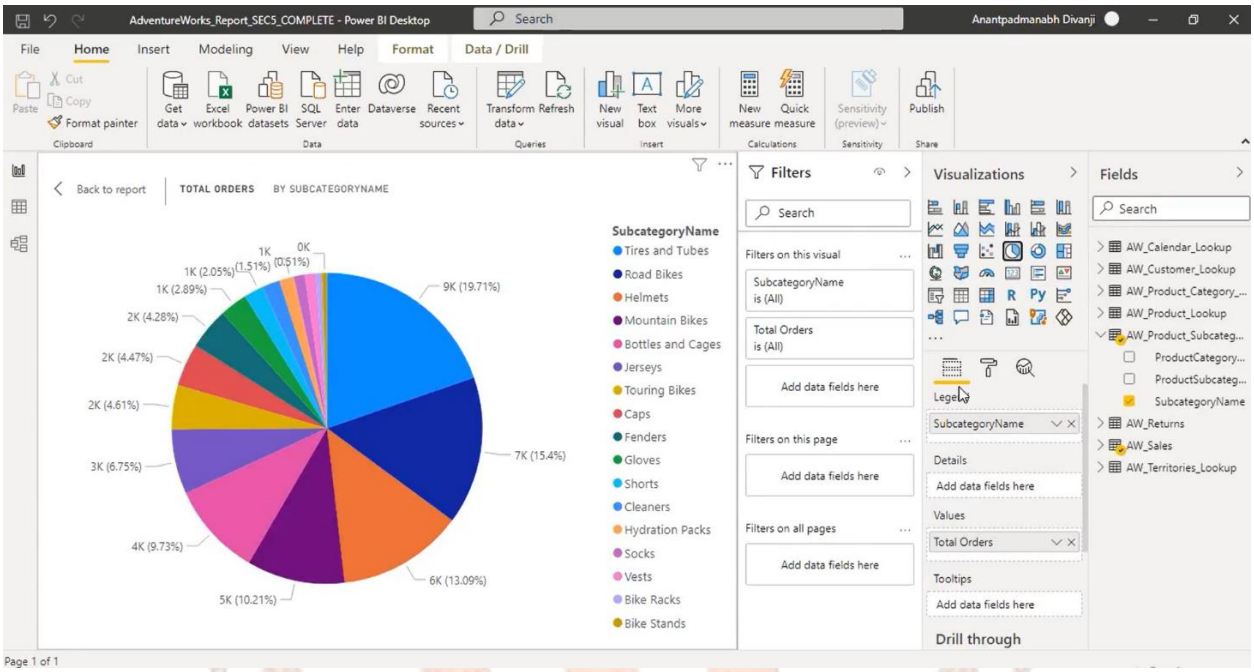


Figure 17: Pie chart

Tooltips are information that appears when the mouse pointer hovers around the visual. Additional content can be added to the tooltip by adding the columns to the respective attributes as shown in Figure 18.

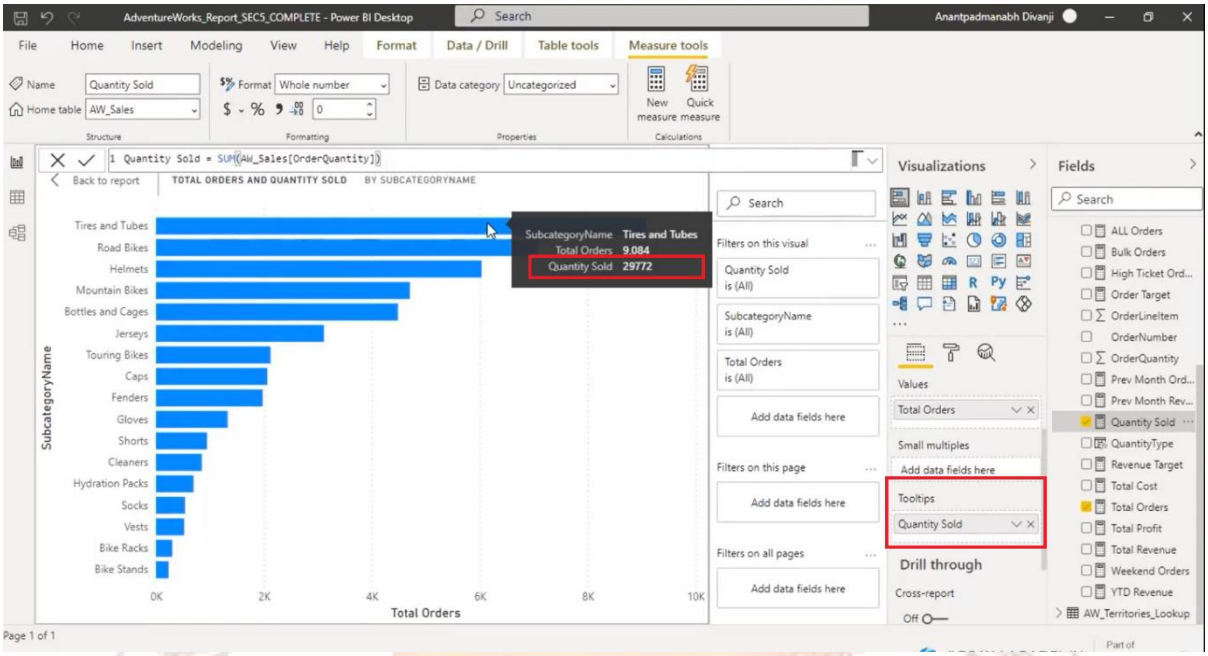


Figure 18: Adding Columns to Tooltips

Additional Formatting can be done by using the 'Format' option as shown in Figure 19 below.

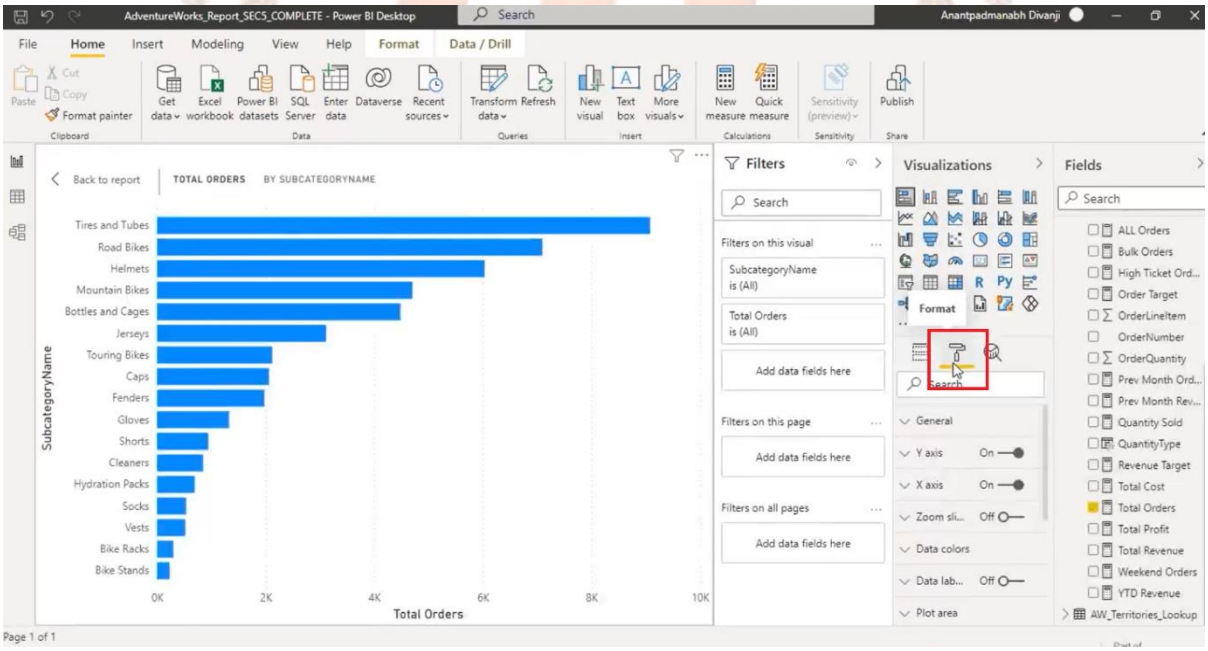


Figure 19: Format Options

There are general options. Also, there are options specific to the X-axis and Y-axis. For example, to change the colour of the content, the 'Color' option of the y-axis can be changed as shown in Figure 20.

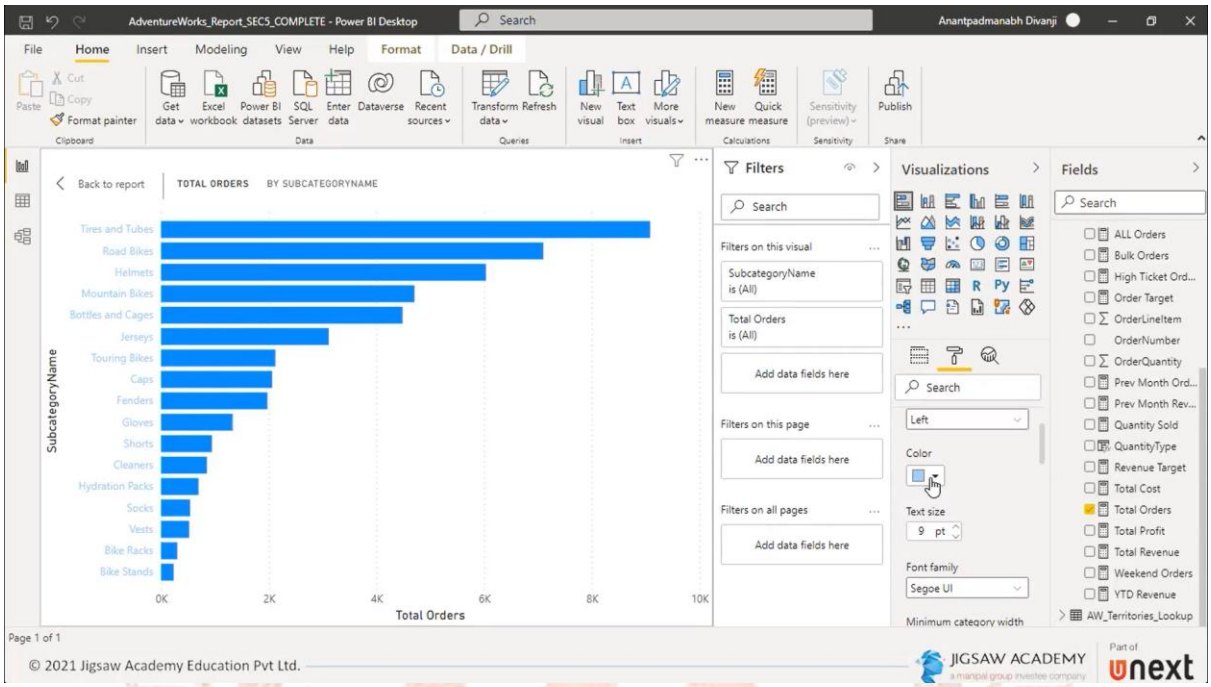


Figure 20: Y-axis specific formatting options

Similarly, alignment, font, size of the bar graph, inner padding between the bar graph and other formatting options can be changed by setting the appropriate attributes in this section. This can be done for both X-axis and Y-axis separately. There is an option to "*Return to Default*" settings after the changes have been made.

There is an option called the *'zoom slider'*. As shown in Figure 21, once the zoom slider is turned on, the slider appears. Using this, the content that needs to appear on the screen can be controlled. Figure 21 displays the bar greater than 5k only. This option can be used to display a range of values also.

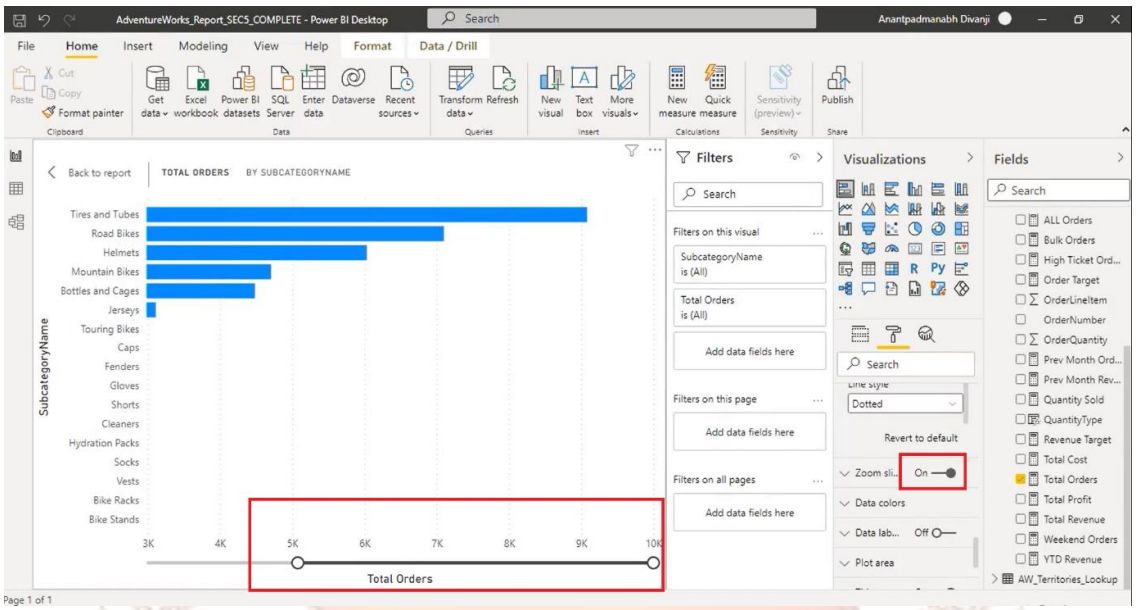


Figure 21: Zoom slider

There are options available to change the colour of the bar chart, and the colour of the text on the bar chart. There are options available to change the Title, the format of the title and so on.

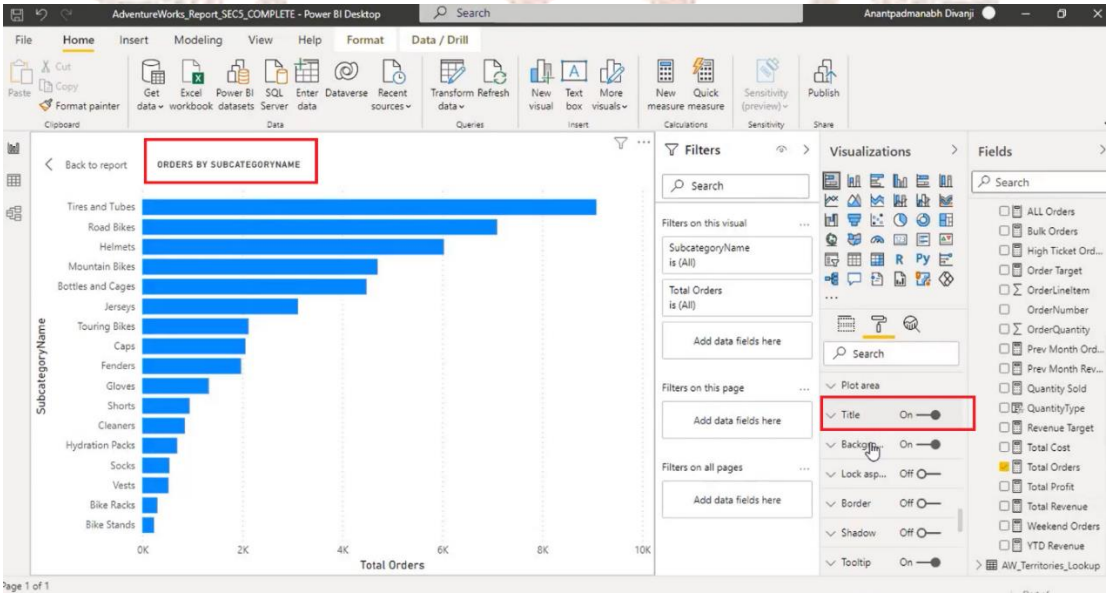


Figure 22: Title Formatting

Similarly, there are formatting options to modify and enhance the background, border and so on.

6. REPORT FILTERING OPTIONS

Power BI has several filtering options that can be used to refine and analyze data. Let's talk about filtering options that are available to us in Power BI report canvas.

Let's add some visuals to the report canvas to explain the filtering options as shown in below figure 23.

Here the user has added two different graphs that have been added from the visualization pane by selecting the required fields.

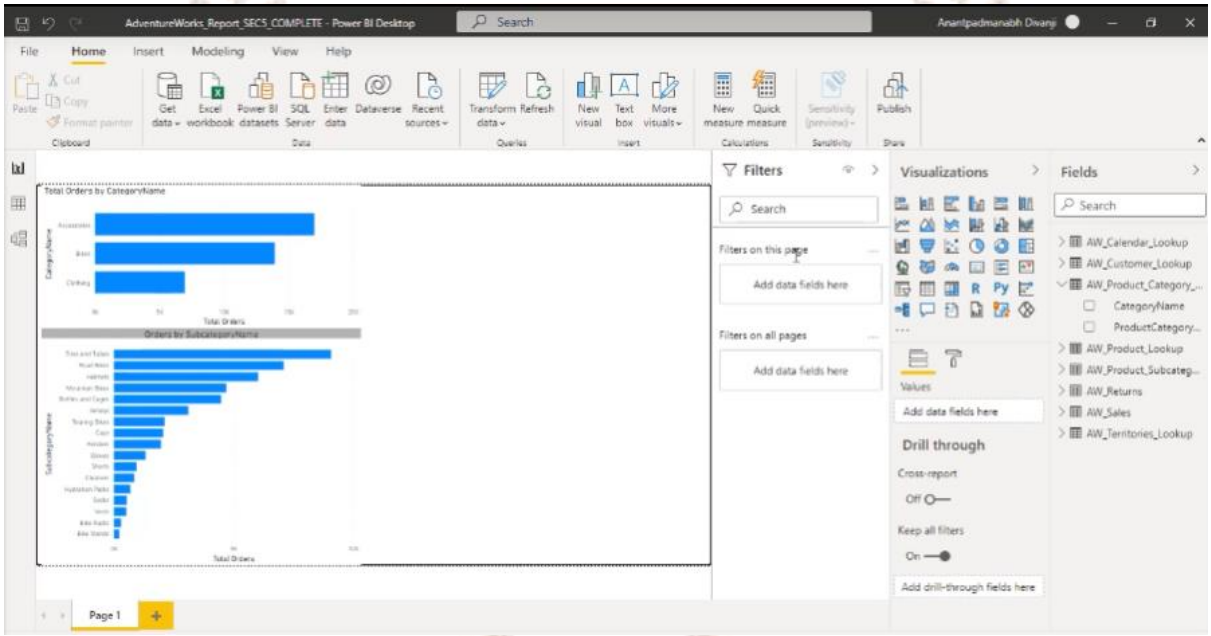


Figure 23: Report Canvas Page with two Visuals

Right now, in filter pane we have two filtering options (i.e., Filters on this Page and Filters on all Pages)

Now, if the user selects a particular visual, then user can notice the options in filters field that can be applied for that particular visual that is selected as shown in figure24.

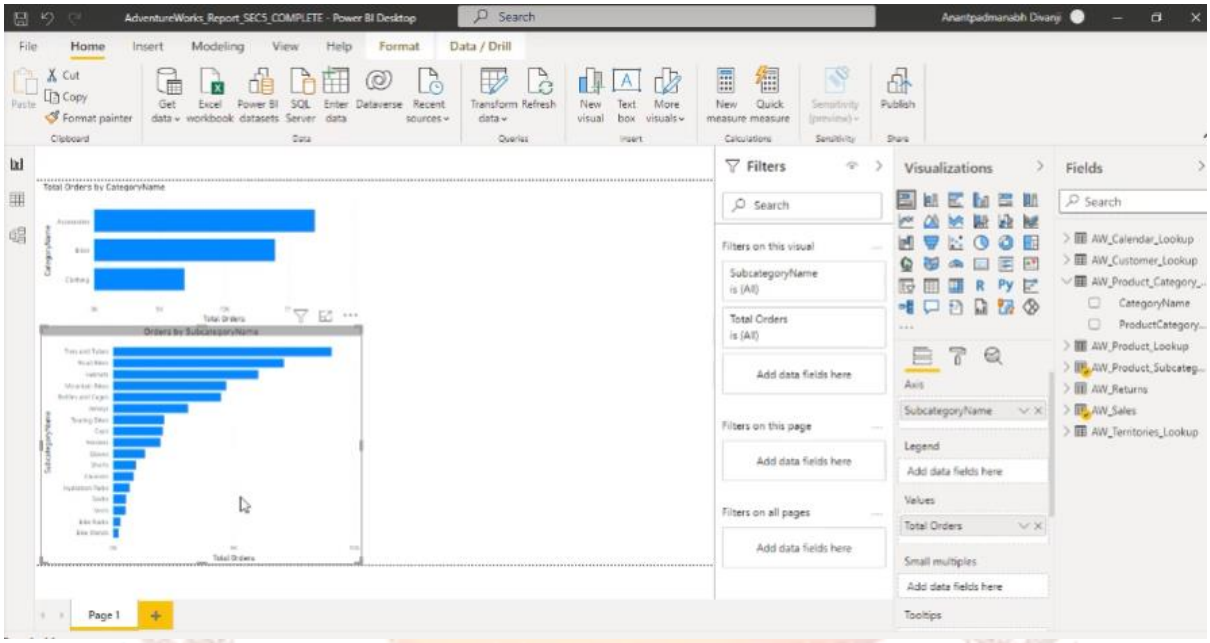


Figure 24: User has selected the visual 1

Now, let us create a new page i.e., page 2, and copy the visual 2 from page 1 and paste it in page 2 and resize the graph if it is required.

Back to page 1, and select the particular bar graph, then user can notice the filters on this visual on *Filters Pane* can be applied only on category name because there are only two attributes are present in that graph i.e., *Category Name and Total Orders* as shown in the below figure 25.

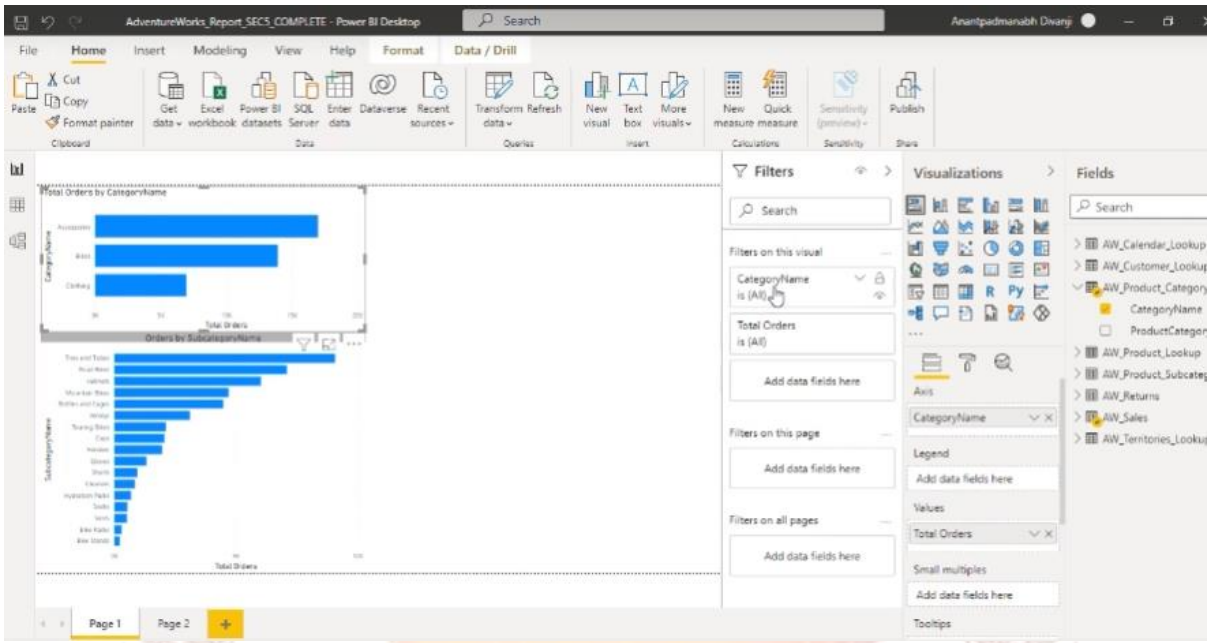


Figure 25: Selected Visual 2 and Filters options have been enabled for the visual.

With respect to category name, if user click on drop down arrow then user will get the *filter type option* and there are two filter type options.

They are, '*Basic Filtering Options*' and '*Advanced Filtering Options'*.

Basic filtering options - where user can select the category on which he wants to visualize the graph.

For example, in the below figure 26 user is selecting Bikes and clothing option and accordingly he will be visualizing the graph.

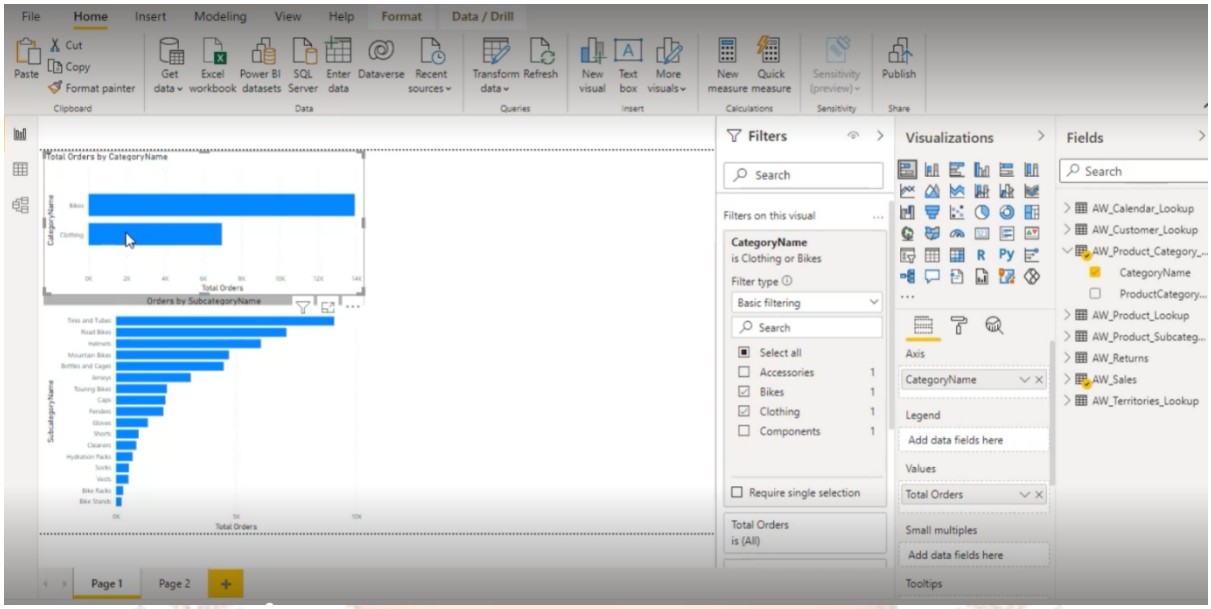


Figure 26: Enabled Basic Filtering Options

Advanced Filtering options – Here if the user wants to filter their data based on whether a specific column contains a certain string of text i.e., by mentioning the string in ***contains*** field, then user can use this option as shown in figure 27.

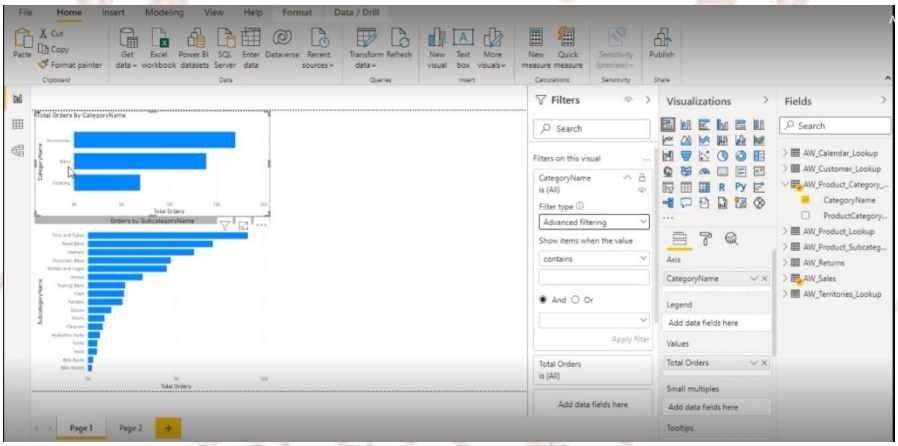


Figure 27: Active page of Advanced Filtering Options

For example, in figure 28, the user has mentioned Bikes in the contains field, and accordingly his graph will be containing only Bike data.

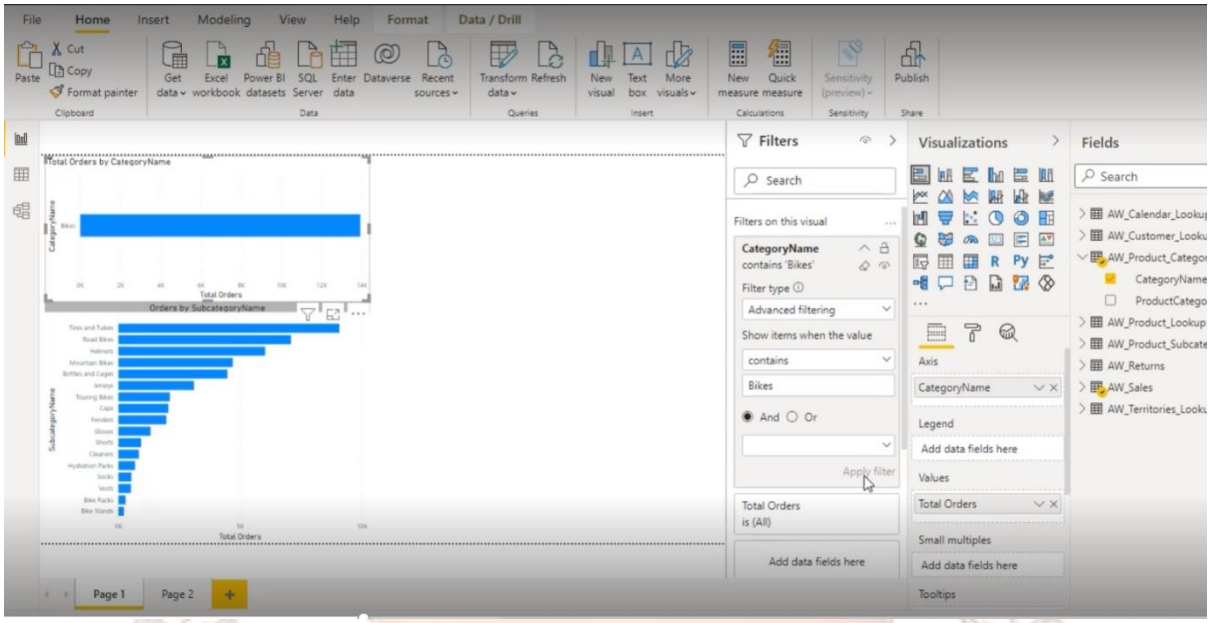


Figure 28: Resultant Graph after applying the Advanced Filtering for 'Bikes'

Apart from *contains*, the advanced filtering has some other options also like 'doesnot', 'is', 'is not', 'starts with' etc as shown in figure 29.

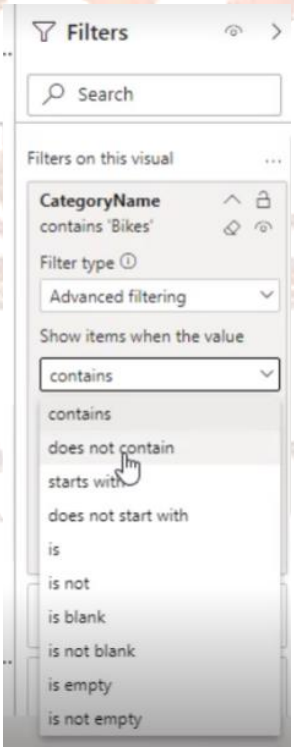


Figure 29: Available options on show items in Advanced Filtering

Apart from this, we have options like "***Top N***" and "***Bottom N***" (In Filter Type field). If user wants to see either top or bottom then user can make use of this option. These options allow users to filter their data to show the top or bottom N items, where N is a specified number.

Let us work on Top N filtering. The below figure 30 shows the Top N filter type, where number of items is to be set (i.e., n value needs to be set) in "*Show items*" field, and accordingly the filter will be applied and it will display visual for Top N items.

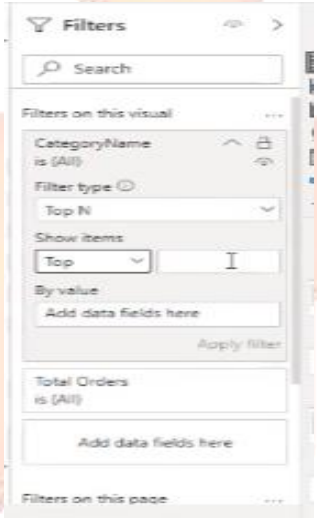
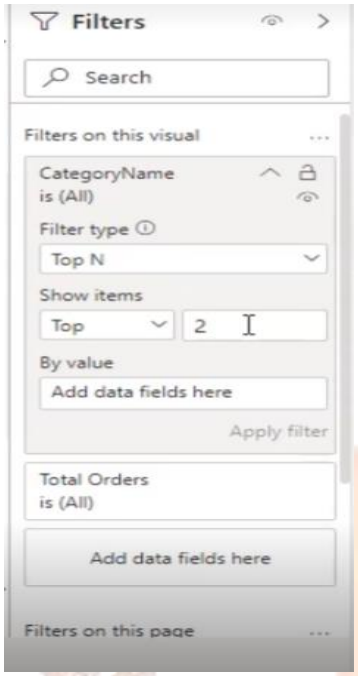


Figure 30: Top N Filtering

In the below figure 31(a) the number of items has been set to 2 (i.e., n=2) and accordingly data fields value to be set in "***By Value***" field (Here we are considering Order Quantity which is selected from Fields pane) and apply filter as shown in figure 31(b).



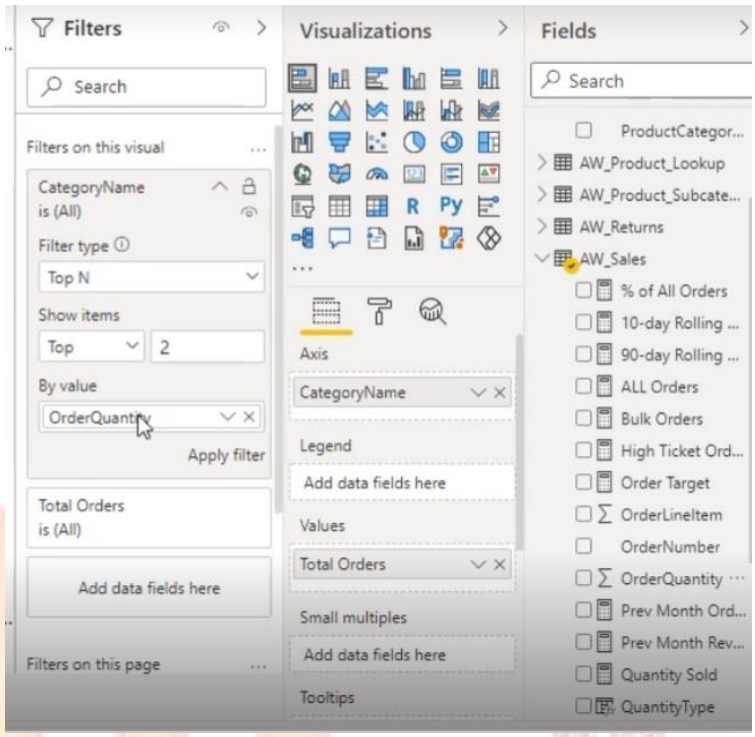


Figure 31(a): N value set to 2 **Figure 31(b): Values set to 'Total Orders'**

After applying filter, user can see the below figure 32 with the graph with top 2 products based on order quantity (i.e., Bikes and Accessories).

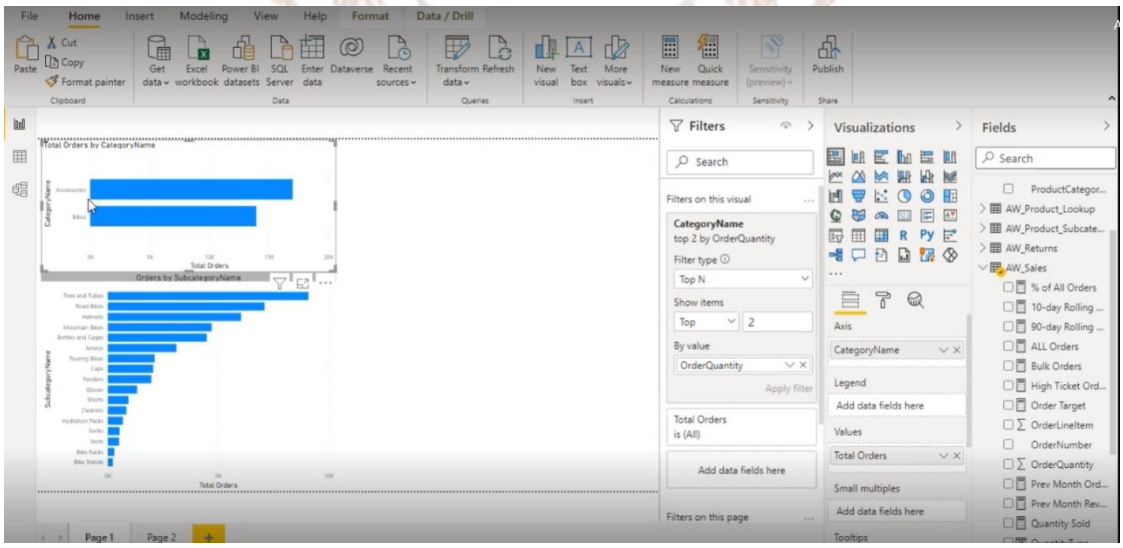


Figure 32: Resultant Graph with Top 2 Products based on order quantity.

Now, let us see another filter (from filters on this visual) called "*Total Orders*", which contains a subfield '*Show items when the value'*.

If the user clicks on '*Show items when the value*' field, then it displays options like is less than, is less than or equal to, is, is not, is greater than etc.as shown in figure 33. (User can select any option based on his requirement)

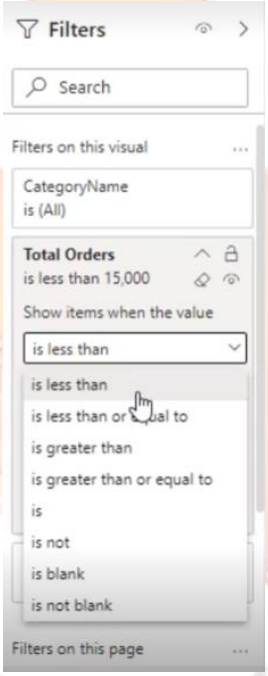


Figure 33: Options of 'Show items when the value'

Next to '*Show items when the value'* we have an empty field to enter the value.

For example, '*Show items when the value*' is set to 'is less than' and the empty field is set to 5000 and click on apply filter then the below graph will be displayed. User can see the empty graph indicating there is no such value where the order is less than 5000 as shown in figure 34.

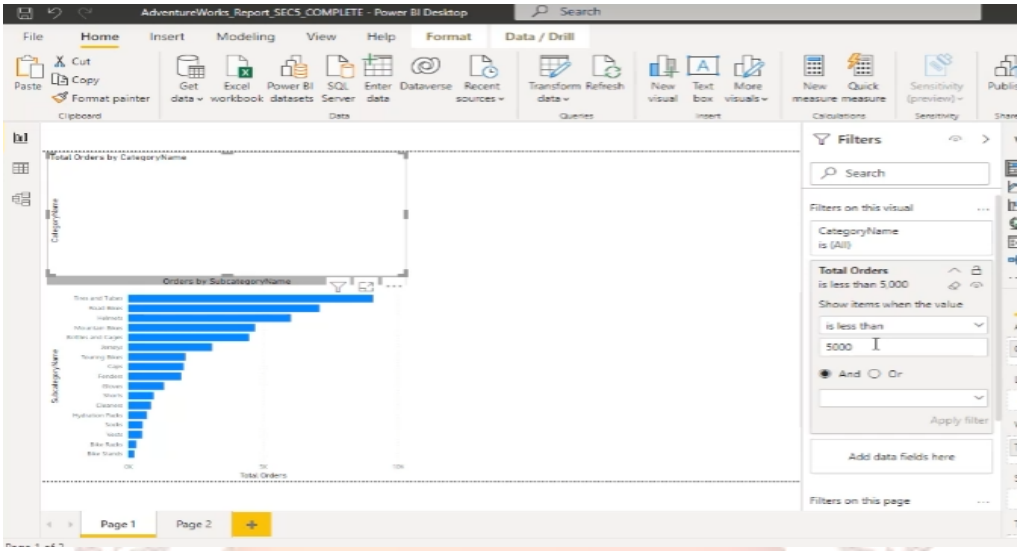


Figure 34: Empty graph indicating there is no such value where the order is less than 5000.

If user enter 15000 (instead of 5000) then the below figure 35 - graph will be displayed indicating that bikes and clothing are ordered less than 15000.

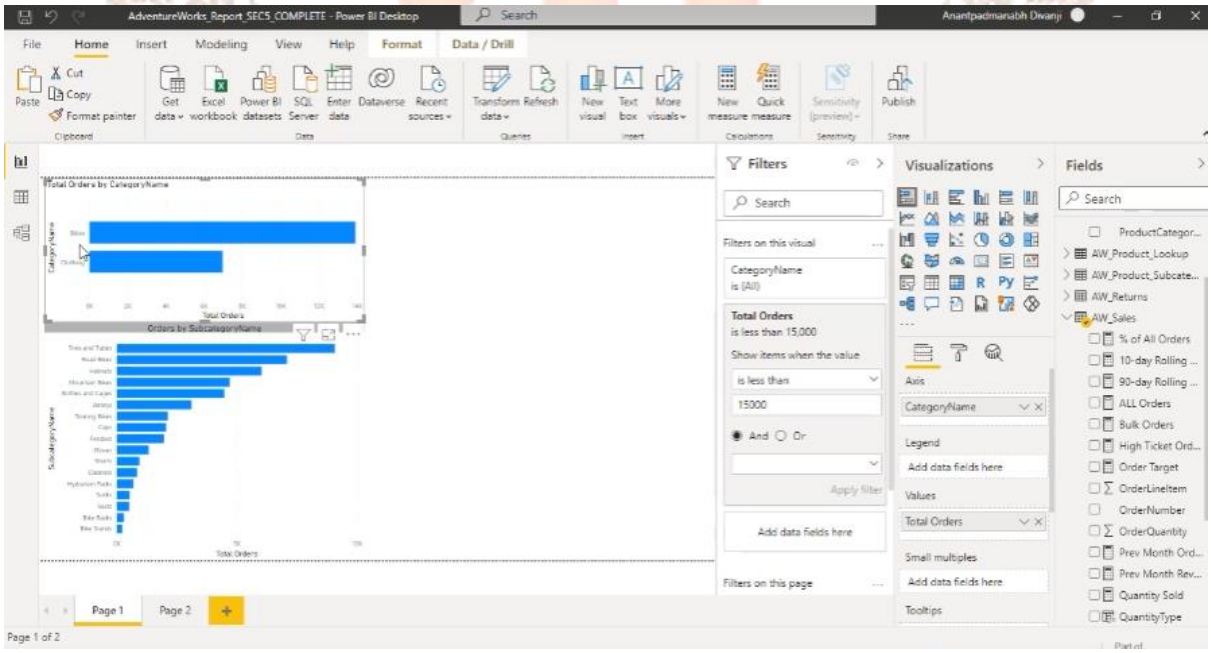


Figure 35: Graphs indicating that bikes and clothing are ordered less than 15000.

Applying Filters to the Entire (this) Page:

Let's consider the user has added the Product Category in the data field from the field section. Then the user can notice the same filter type as discussed before i.e., basic filtering and advanced filtering.

Here if we select the options let say bike, then the click on apply. Then changes have been applied to all the visuals (graphs) that are present on that page as shown in figure 36.

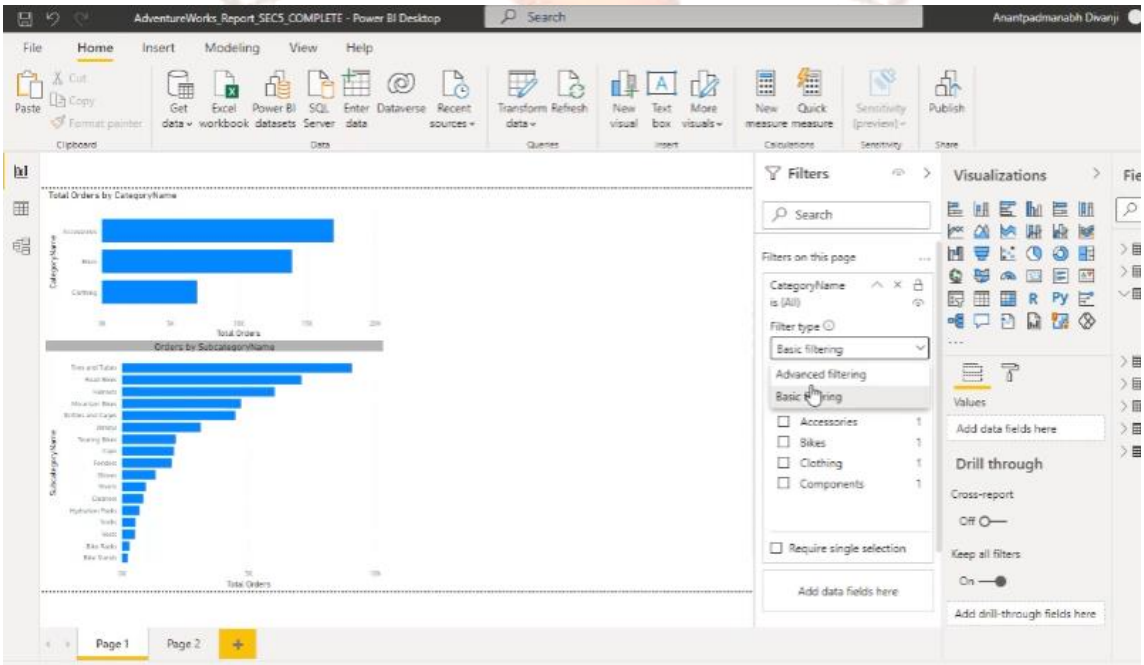


Figure36 (a): Before applying the filter of 'Filters on this Page'

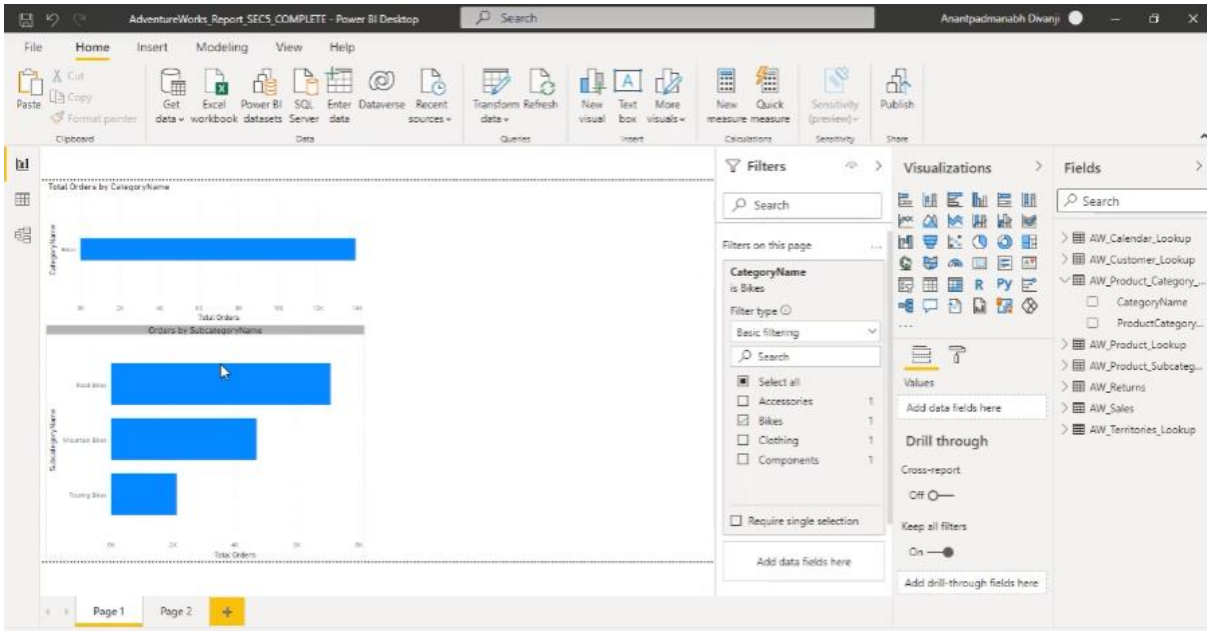


Figure 36 (b): After applying the filter of 'Filters on this Page'.

Similarly, we can apply advanced Filtering by selecting the options from '*Show items when the value*' and by entering the value in the next field.

For example, let's select the 'or' operation for 'contains' field to visualize either Bikes or Clothing data:

Filter Type: Advanced Filtering

Show items when the value: contains

Bikes (to be entered in the empty field after the item field) and check the '*or*' option

Again, select the 'contains' and the second category after that i.e., clothing and apply the filter.

The result is shown in figure 37, where the filter has been applied to both visuals.

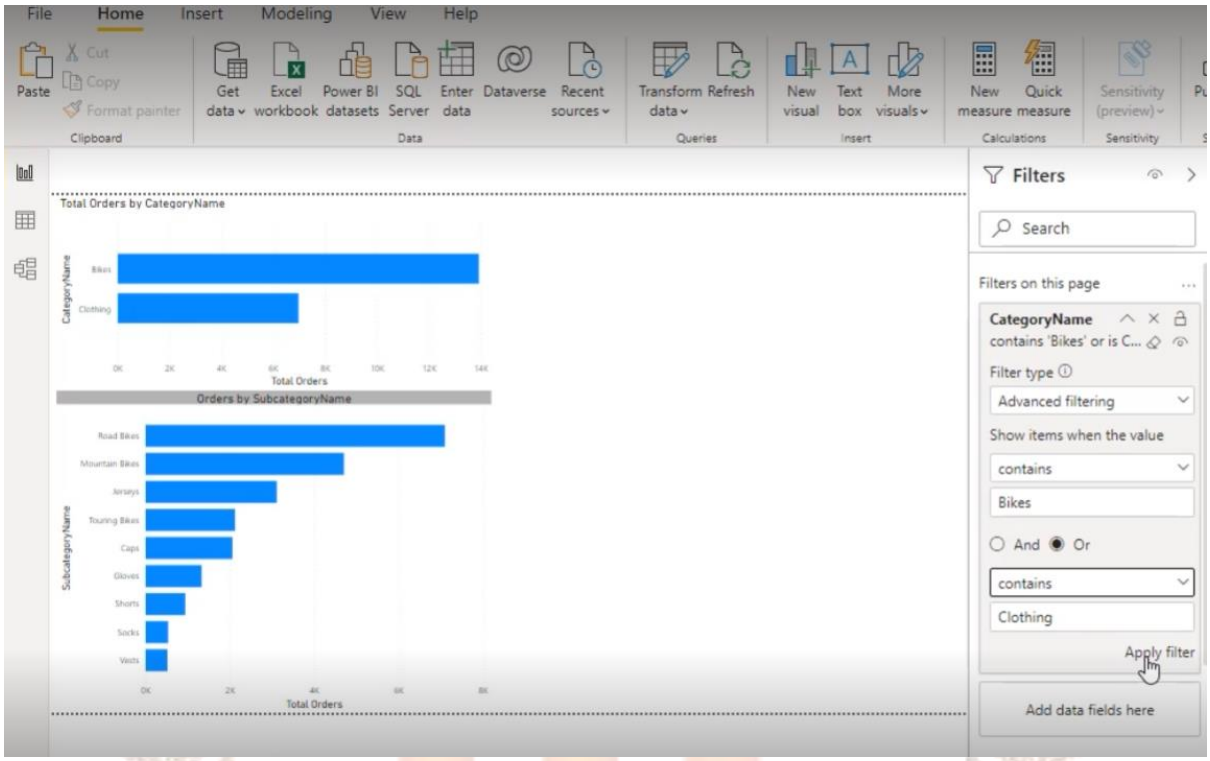


Figure 37: Resultant visuals after applying advanced filtering

Now, look back at the page 2, where no changes have been applied to the visuals (as we have applied changes for the current entire page and not for all the pages).If user wants to check the page 2 then clear the filters (by clicking on eraser icon which is present near category name field) and select Filters on all pages.

Applying Filters on all pages:

Select the Filters on all pages and select the 'category name' from the 'Fields'. Again, here we have Basic filtering and Advanced filtering.

Let's work with Basic Filtering, and select the clothing and Bikes from the options. Then in all the pages (here we have Page 1 and Page 2) we can see the updating in visuals with respect to clothing and Bikes only as shown in figure 38(a) and (b).

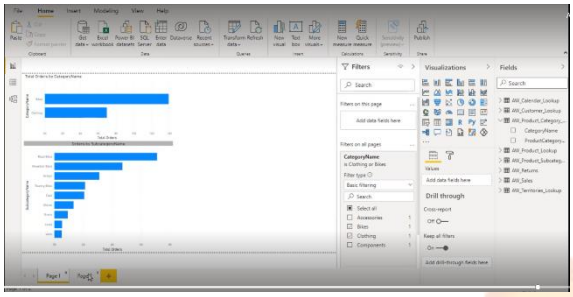


Figure 38(a): Applied filter to page 1

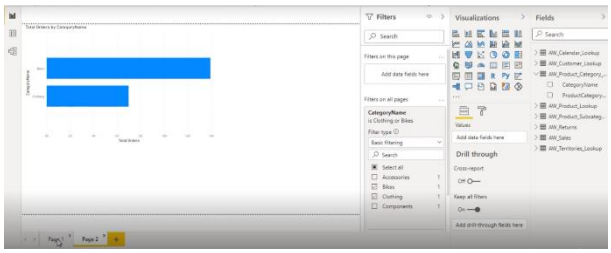


Figure 38(b): Applied filter to page 2.

Next will see how to highlight the particular attribute.

Let's clear all the filters and work on Page 1. In this page if user selects Bikes by clicking on the bar graph as shown in the below figure 39, then Bike bar will be highlighted and the corresponding sub categories from the second graph will be highlighted indicating that this is also a filtering option.

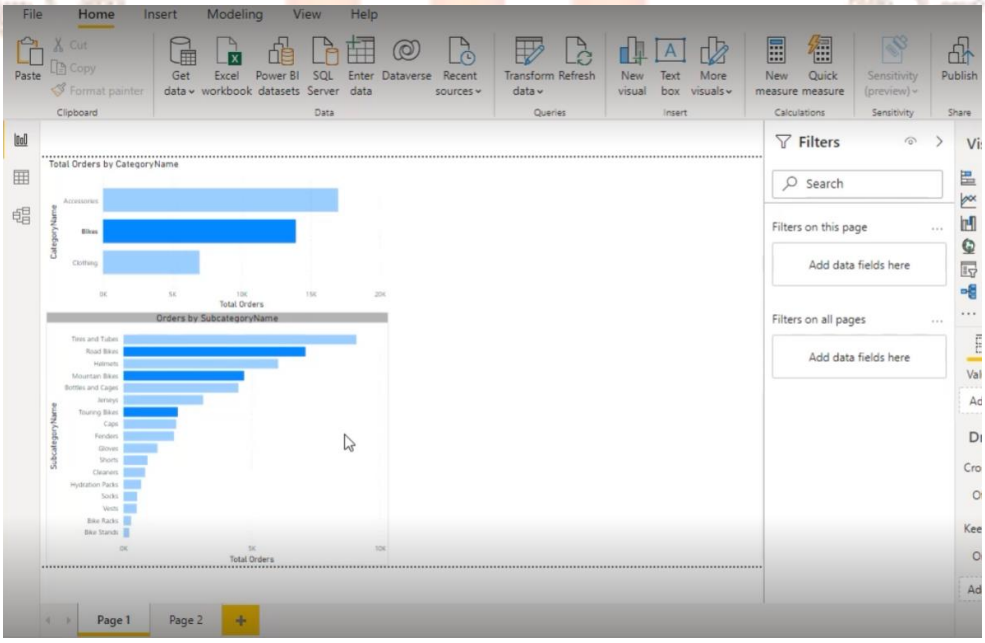


Figure 39: Highlighting the Particular category and sub category in the graph.

7. REPORT FORMATTING OPTIONS

Power BI is a well-known application for data visualization and reporting that gives you a variety of formatting options to help you produce polished and interesting reports.

Now, let's understand the report formatting options by considering a sample graph as shown in figure 40.

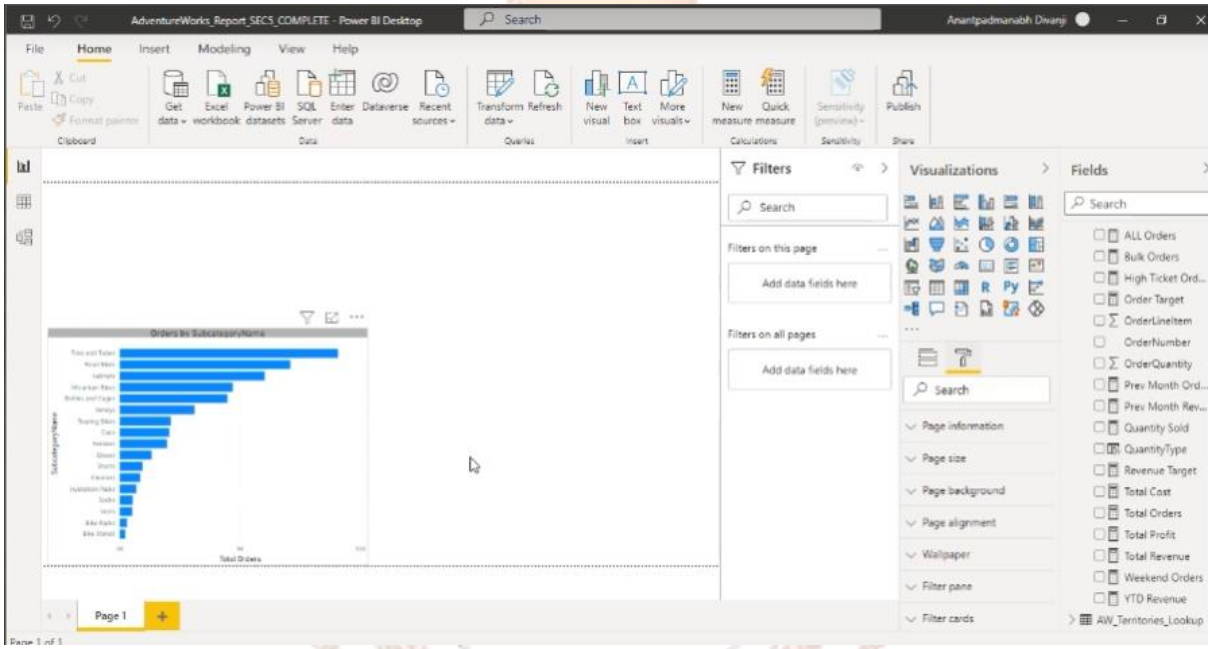


Figure 40: Sample Report Canvas with a visual.

To change the view of the displayed visual, then will perform the below steps:

1. Select the Visual (bar graph) by clicking on it, so that it gets highlighted. As soon as the visual gets highlighted, the fields are updated under the field pane indicating what are all the fields present in the visual as shown in figure 41.

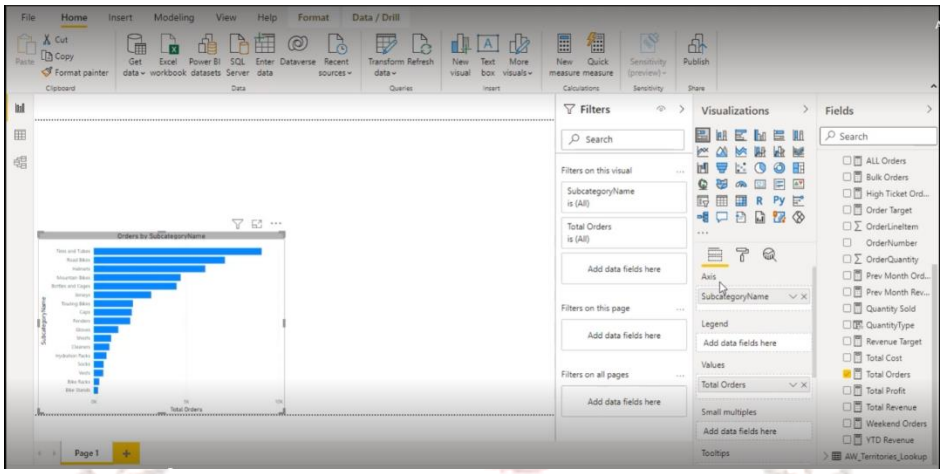


Figure 41: Highlighted Fields in 'Field Pane'

If the user clicks on formatting options, then formatting options also get updated concerning bar graphs like the x-axis, y-axis etc. as shown in figure 42.

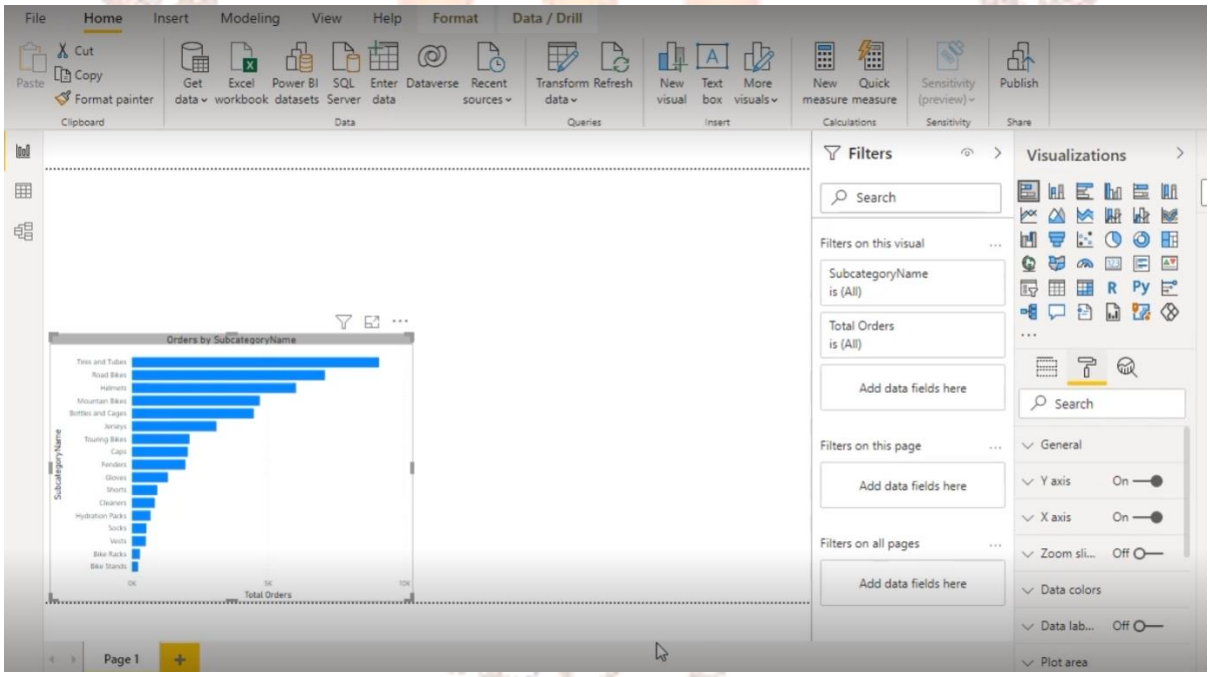


Figure 42: Updated formatting options in 'Formatting Pane'

If a user wants to change the visual of the graph, let's say instead of a Bar graph (shown in fig 42) user wants to view it as a pie chart. The user has to click the pie-chart symbol from the visualization pane and update the graph. As soon as the graph is updated the user can notice that formatting options will also get updated with options like Legend, Data colors, Title etc. The updated pie chart and formatting options can be seen in figure 43.

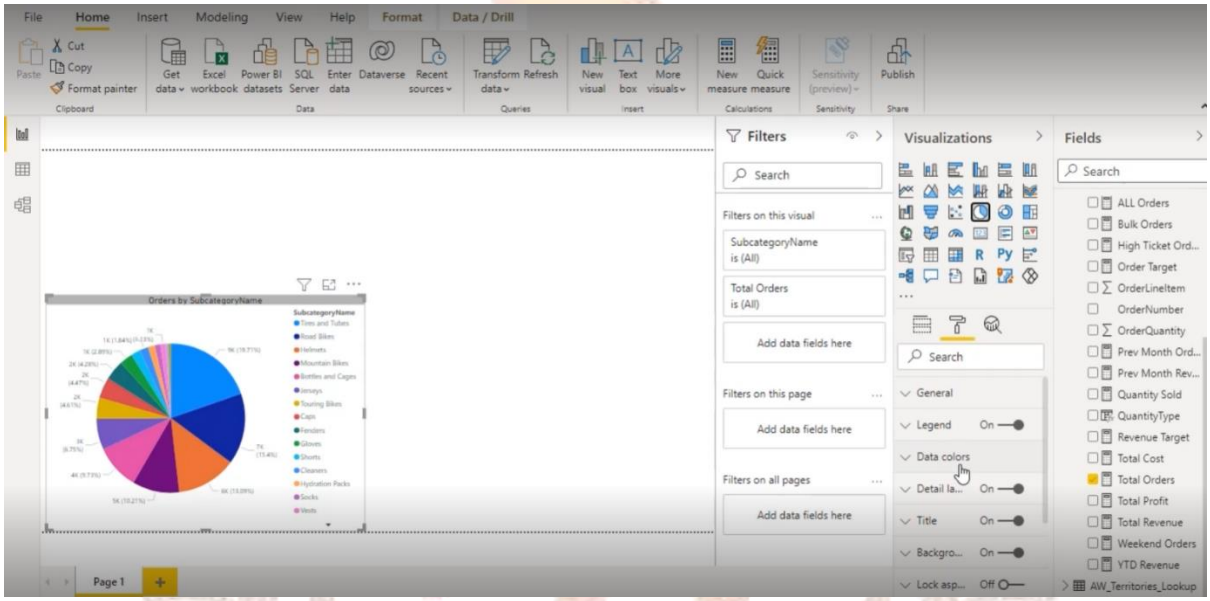


Figure 43: Bar graph converted to the pie chart with updated formatting options.

Basically, the field pane and the formatting options are specific to the chart type the user selects to visualize the data.

For example, if the user wants to view data as a line graph then according to that the field pane/section and formatting options will be updated as shown in figure 44.

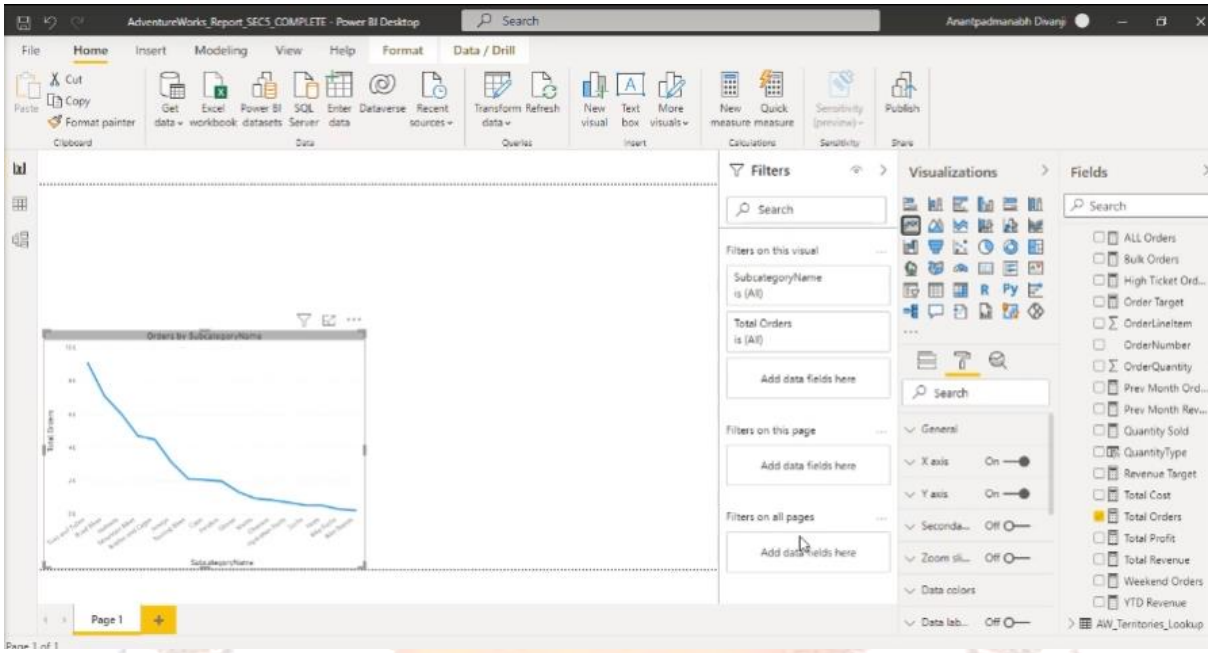


Figure 44: Visualizing Data as a line graph with updated formatting options view

Self-Assessment Questions -1

1. Name some of the information displayed in the startup or the launch page
2. The _____in Power BI is the main work area where users can create and design reports and visualizations using data from various sources.
3. The default chart selected for a integer column with continuous values is a _____.
4. To display a horizontal bar chart per subcategory, set the subcategory column to the _____ attribute.
5. To control the range of values that appear on the screen the _____option can be enabled.
6. The _____ in Power BI is where you design and build the visualizations and reports that you want to present to your audience.
7. Visualizations can be moved or resized with _____ enabled, and the visualizations will automatically align to the closest grid line.
8. The actual data sets or data tables will appear in _____ section
9. User can customize which data to be shown in his visualizations by using _____
10. Numbers appearing in Toolkit can be formatted to include symbols like ',', '%' etc -True or False

8. TERMINAL QUESTIONS

1. What are the different types of visualizations available in Power BI and how can a user format them?
2. Explain the procedure to apply conditional formatting to the report visuals in Power BI?
3. Elucidate the procedure to apply a filter to a specific visual or group of visuals in Power BI?
4. How can a user use a filter to highlight specific data in a Power BI report?
5. Briefly explain the key features and components of the report view.

9. SELF-ASSESSMENT QUESTIONS – ANSWERS

1. Recent Sources connected, Getting Started Videos, Updates, Blogs, Forums, Tutorials etc
2. report canvas
3. vertical bar chart.
4. "Axis"
5. "zoom slider"
6. Report View
7. Snap-To-Grid
8. Field
9. Filters
10. True

Terminal Questions – Answers

1. Refer section 'Basic Charts and Conditional Formatting'
2. Refer section 'Basic Charts and Conditional Formatting'
3. Refer section 'Report Filtering Options'
4. Refer section 'Report Filtering Options'
5. Refer section 'Exploring the Report Canvas'